



# Zeta potential of artificial and natural calcite in aqueous solution



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## ABSTRACT

Despite the broad range of interest and applications, controls on calcite surface charge in aqueous solution, especially at conditions relevant to natural systems, remain poorly understood. The primary data source to understand calcite surface charge comprises measurements of zeta potential. Here we collate and review previous measurements of zeta potential on natural and artificial calcite and carbonate as a resource for future studies, compare and contrast the results of these studies to determine key controls on zeta potential and where uncertainties remain, and report new measurements of zeta potential relevant to natural subsurface systems.

The results show that the potential determining ions (PDIs) for the carbonate mineral surface are the lattice ions  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{CO}_3^{2-}$ . The zeta potential is controlled by the concentration-dependent adsorption of these ions within the Stern layer, primarily at the Outer Helmholtz Plane (OHP). Given this, the Iso-Electric Point (IEP) at which the zeta potential is zero should be expressed as pCa (or pMg). It should not be reported as pH, similar to most metal oxides.

The pH does not directly control the zeta potential. Varying the pH whilst holding pCa constant yields constant zeta potential. The pH affects the zeta potential only by moderating the equilibrium pCa for a given  $\text{CO}_2$  partial pressure ( $p\text{CO}_2$ ). Experimental studies that appear to yield a systematic relationship between pH and zeta potential are most likely observing the relationship between pCa and zeta potential, with pCa responding to the change in pH. New data presented here show a consistent linear relationship between equilibrium pH and equilibrium pCa or pMg irrespective of sample used or solution ionic strength. The surface charge of calcite is weakly dependent on pH, through protonation and deprotonation reactions that occur within a hydrolysis layer immediately adjacent to the mineral surface. The Point of Zero Charge (PZC) at which the surface charge is zero could be expressed as pH, but surface complexation models suggest the surface is negatively charged over the pH range 5.5–11.

Several studies have suggested that  $\text{SO}_4^{2-}$  is also a PDI for the calcite surface, but new data presented here indicate that the value of  $p\text{SO}_4$  may affect zeta potential only by moderating the equilibrium pCa. Natural carbonate typically yields a more negative zeta potential than synthetic calcite, most likely due to the presence of impurities including clays, organic matter, apatite, anhydrite or quartz, that yield a more negative zeta potential than pure calcite. New data presented here show that apparently identical natural carbonates display differing zeta potential behaviour, most likely due to the presence of small volumes of these impurities. It is important to ensure equilibrium, defined in terms of the concentration of PDIs, has been reached prior to taking measurements. Inconsistent values of zeta potential obtained in some studies may reflect a lack of equilibration.

The data collated and reported here have broad application in engineering processes including the manufacture of paper and cement, the geologic storage of nuclear waste and  $\text{CO}_2$ , and the production of oil and gas.

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## 1. Introduction

The properties of the calcite mineral surface and the interface between calcite and aqueous solution are of broad interest in many areas of science and engineering. Calcite is a common mineral, comprising approximately 4% of the Earth's crust, and surface reactions on calcite play an important role in many geochemical and environmental systems, as well as many areas of industry, including the manufacture of paper and cement (e.g. [5]), the geologic storage of nuclear waste and CO<sub>2</sub> (e.g. [30]), and the production of oil and gas (e.g. [9,28,86,88]). However, despite the broad range of interest and applications, controls on calcite surface charge in aqueous solution, especially at conditions relevant to natural systems, remain poorly understood. Numerous papers have reported inconsistent or contradictory data and models, and there is still active debate over the relationship between calcite surface charge and electrolyte pH, the concentration of ions such as Ca<sup>2+</sup>, Mg<sup>2+</sup>, CO<sub>3</sub><sup>2−</sup> and SO<sub>4</sub><sup>2−</sup> in aqueous solution, the partial pressure of CO<sub>2</sub>, the difference between natural and artificial calcite, and the role of dissolution and/or precipitation.

The focus of this paper is the zeta potential of artificial and natural calcite in aqueous solution. The zeta potential is an important measure of the electrical potential at the mineral surface, and the magnitude and sign of the zeta potential control the electrostatic interactions between the mineral surface and polar species in aqueous solution, between the mineral surface and other charged interfaces such as the water–air and water–oil interfaces, and between mineral particles in suspension including flocculation and dispersion. Measurements of zeta potential in low concentration solutions and at laboratory conditions are relatively straightforward, and most studies of the calcite mineral surface have reported measurements of zeta potential or the closely related property of electrophoretic mobility. Other approaches to determine surface charge, such as potentiometric titration, are challenging to apply in calcite because rapid dissolution kinetics and the buffering effect of carbonate ions in solution can affect the results. Thus the primary data source to understand calcite surface charge comprises

measurements of zeta potential. Where relevant, we use the results of spectroscopic, microscopic, surface diffraction, modelling and theoretical studies to help explain experimentally determined values of zeta potential. However, a comprehensive review of these studies is beyond the scope of the paper; a companion paper of comparable length would be required to do justice to this work.

The aims of the paper are therefore threefold; (1) to collate and review previous measurements of zeta potential on natural and artificial calcite and carbonate as a resource for future studies, (2) to compare and contrast the results of these studies to determine key controls on zeta potential and where uncertainties remain, and (3) to report new measurements of zeta potential relevant to natural subsurface systems. There has been no comprehensive review of zeta potential measurements in calcite and natural carbonate to date, although Wolthers et al. [85] collated published zeta potential data to constrain their new surface complexation model. Moreover, there are a lack of data which can be applied to natural systems owing to the comparatively high ionic strength (typically >0.01 M and often >2 M) and complex compositions (including Na<sup>+</sup>, Ca<sup>2+</sup>, Mg<sup>2+</sup>, Cl<sup>−</sup>, CO<sub>3</sub><sup>2−</sup>, SO<sub>4</sub><sup>2−</sup>) of natural brines, compared to the simple, dilute aqueous solutions typically used in laboratory experiments.

We report new data obtained using (intact) natural samples saturated with electrolytes relevant to natural systems and an experimental methodology specifically designed to allow this parameter space to be explored. Data obtained in this parameter space are still very scarce, despite the broad range of interest. Our results demonstrate that apparently identical natural carbonates can exhibit differing zeta potential behaviour, suggesting that small variations in the type and content of impurities in the mineral lattice can significantly impact the surface charge. However, the effect of the lattice ions Ca<sup>2+</sup> and Mg<sup>2+</sup> on the zeta potential of a given carbonate is identical within experimental error. The presence of apparently indifferent ions such as Na<sup>+</sup> and Cl<sup>−</sup> at the high concentrations typical of natural brines can significantly shift the isoelectric point expressed as pCa and pMg. We demonstrate experimentally that equilibrium pH is strongly correlated to

equilibrium pCa and pMg, and apparent trends in zeta potential with pH reflect trends in zeta potential with pCa and pMg.  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  are the potential determining ions (PDIs) for the calcite surface. We provide experimental evidence suggesting that  $\text{SO}_4^{2-}$ , which has previously been suggested as a PDI, may not directly control the zeta potential of calcite; rather, varying  $\text{pSO}_4$  causes variations in pCa which can be correlated with observed changes on zeta potential. These new results have broad application to subsurface engineering processes such as the geologic storage of nuclear waste and  $\text{CO}_2$ , and the production of oil and gas.

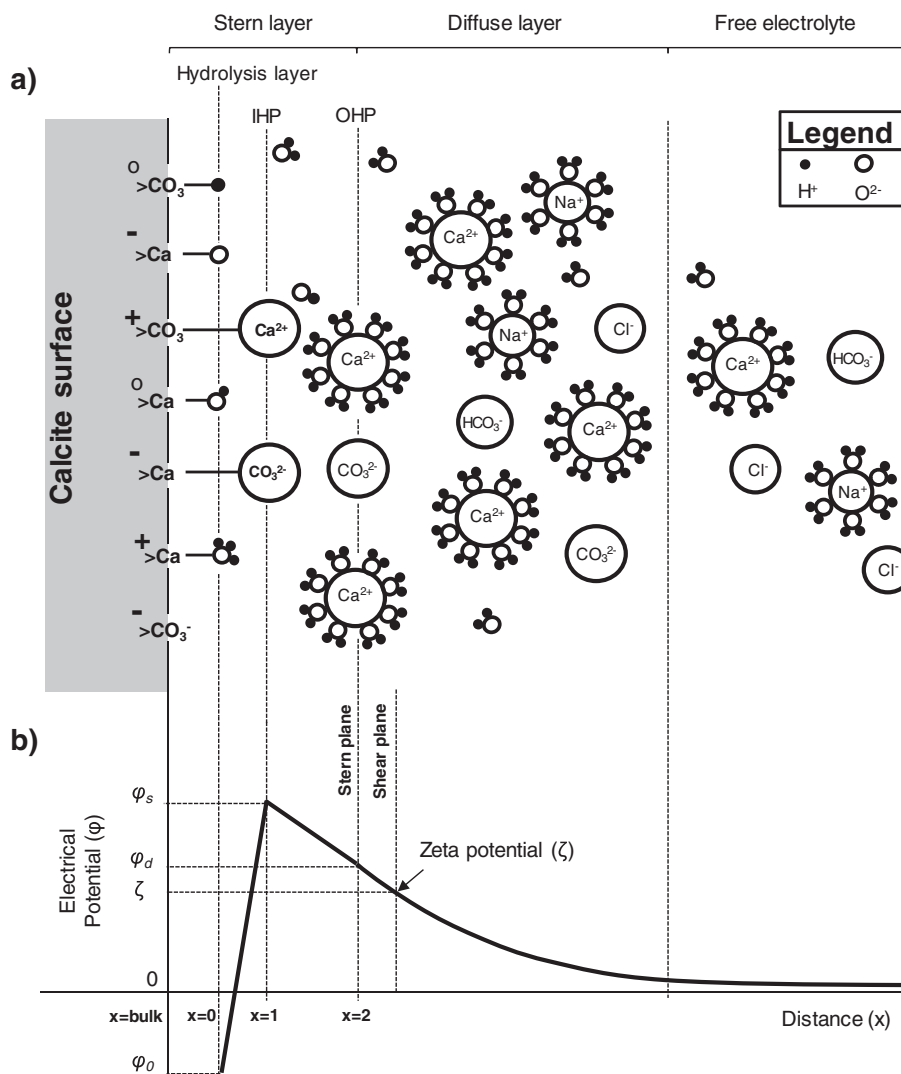
## 2. Zeta potential and the electrical double layer

### 2.1. The electrical double layer

The immersion of a calcite mineral in aqueous solution leads to a separation of electrical charge at the mineral–solution interface. An excess of charge at the mineral surface is balanced by a region of equal but opposite charge in the adjacent solution, in an arrangement often termed the electrical double layer (EDL; e.g. [25,26]; see Fig. 1). The charge at the mineral surface is balanced by a relative decrease in the concentration of co-ions (i.e. ions with the same charge as the surface)

and increase in the concentration of counter-ions (i.e. ions with the opposite charge as the surface) in the adjacent solution. The region immediately adjacent to the mineral surface is typically termed the Stern layer, and contains ions that are attached to the mineral surface; the Stern layer may be further divided into the inner and outer Helmholtz layers, defined by the inner and outer Helmholtz planes (the IHP and OHP shown in Fig. 1a; note the inner and outer Helmholtz layers and associated planes are also sometimes termed the inner and outer Stern layers and planes). The IHP defines the location of ions that closely approach the mineral surface and are attached to surface sites (Fig. 1a). The OHP defines the location of larger, typically hydrated ions that cannot enter the inner Helmholtz plane but which are nevertheless attached to the mineral surface [6].

In most cases, the charge in the Stern layer does not exactly balance the surface charge, giving rise to a 'diffuse' or 'Gouy–Chapman' layer that contains the remaining excess charge in the solution required to ensure electrical neutrality of the EDL (e.g. [43]). The difference between the Stern and diffuse layers is that the co- and counter-ions in the diffuse layer are not attached to the mineral surface. At low ionic strength ( $\leq 0.1\text{M}$ ), the ion concentration in the diffuse layer decreases exponentially with distance from the OHP (Fig. 1a). The electrical



**Fig. 1.** (a) A schematic representation of the electrical double layer at the calcite–water interface. The surface speciation sites ( $>\text{CO}_3\text{H}^0$ ,  $>\text{CaOH}^+$ ,  $>\text{CO}_3\text{Ca}^+$ ,  $>\text{CaOH}^0$ ,  $>\text{CaCO}_3^-$ ,  $\text{CaOH}_2^+$  and  $>\text{CO}_3^-$ ) are from [57]. The Stern layer is described by three planes. The 0-plane ( $x = 0$ ) corresponds to the hydrolysis layer where H and OH are chemi-bonded to the bulk ions [71]. The 1-plane ( $x = 1$ ) denotes inner-sphere complexes and corresponds to the inner Helmholtz plane (IHP), while the 2-plane ( $x = 2$ ) denotes outer-sphere complexes and corresponds to the outer Helmholtz plane (OHP; [85]). (b) A schematic representation of the variation in the electrical potential with distance from the mineral surface. Here the mineral surface is negatively charged, consistent with the surface complexation model of [29], but the zeta potential is positive because of adsorption of the lattice PDIs  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$ .

potential corresponding to the charge distribution within the EDL decreases linearly with distance from the mineral surface through the Stern layer, although there may be a difference in gradient between the inner and outer Helmholtz layers, and exponentially with distance from the OHP through the diffuse layer, falling to zero in the uncharged solution (Fig. 1b).

## 2.2. Electrokinetic phenomena

Electrokinetic phenomena arise when there is a relative motion between the excess charge in the diffuse layer of the EDL and the charged surface [34]. It is typically assumed that the excess charge in the diffuse layer is mobile only beyond a plane termed the 'shear' or 'slip' plane that lies a small distance away from the OHP (Fig. 1). The zeta potential is the electrical potential at the shear plane. In some models of the EDL, it is assumed that the shear plane and the OHP are identical, in which case the zeta potential also represents the electrical potential at the OHP. Methods to determine the zeta potential make use of electrokinetic phenomena, most typically (1) electrophoresis, which is the mobilisation of charged particles, relative to a stationary solution, under the influence of an applied electrical field, and (2) streaming potential, which is the potential difference that arises when a solution is moved relative to a stationary solid under the influence of an applied pressure gradient [15].

The electrophoretic method (termed here EPM) used to determine zeta potential is conducted on powdered samples suspended in the solution of interest. An electrical field  $E$  is applied across the suspension and the resulting velocity  $v_e$  of the charged particles is measured with respect to the solution. The velocity is then normalized by the electrical field to yield the electrophoretic mobility  $u_e$ , which is typically treated as an isotropic quantity that can be related to the zeta potential by the Helmholtz–Smoluchowski equation for electrophoresis [15]

$$u_e = \frac{\varepsilon \zeta}{\mu}, \quad (1)$$

where  $\varepsilon$  is the permittivity ( $\text{F m}^{-1}$ ) and  $\mu$  the viscosity ( $\text{Pa s}$ ) of the solution. The zeta potential obtained using the EPM is an effective value across all the suspended particles; the zeta potential of individual particles (and, indeed, between the different faces of a given particle) may differ from the single value interpreted from EPM measurements.

The streaming potential method (termed here SPM) used to determine zeta potential is conducted using intact samples of porous materials, packed beds of particles, or surfaces through, or across, which the solution of interest is caused to flow. A pressure difference  $\Delta P$  is applied across the sample and the resulting electrical potential difference  $\Delta V$  is measured. The potential difference is then normalized by the pressure difference to yield the streaming potential coupling coefficient, which is typically treated as an isotropic quantity that can be related to the zeta potential by the Helmholtz–Smoluchowski equation for streaming potential (e.g. [17,34,52])

$$C = \frac{\varepsilon \zeta}{\sigma_s \mu}, \quad (2)$$

where  $\sigma_s$  is the electrical conductivity of the solution of interest ( $\text{S m}^{-1}$ ). When streaming potential measurements are obtained in porous media, it may be necessary to use a version of Eq. (2) modified to account for the effect of enhanced electrical conductivity through the EDL (often termed surface electrical conductivity; [49])

$$C = \frac{\varepsilon \zeta}{\sigma_{ss} \mu F}, \quad (3)$$

where  $\sigma_{ss}$  is the electrical conductivity of the system saturated with the solution of interest and  $F$  is the so-called formation factor (dimensionless) which is defined as the ratio of the solution conductivity  $\sigma_s$  to

the system conductivity  $\sigma_{ss}$  when surface electrical conductivity is negligible. The formation factor is typically measured on the system saturated with a concentrated solution with high electrical conductivity (e.g. [35,40,83]).

## 3. Development of surface charge on calcite in aqueous solution

### 3.1. Effect of pH

The development of surface charge on calcite in aqueous solution is still not fully understood [71,79,85]. In most metal oxides, charge development occurs when mineral surfaces sites are hydrated to form amphoteric groups  $>\text{MeOH}$  (where Me denotes a metal ion and  $>$  denotes the crystal lattice) and these react either with  $\text{H}^+$  or  $\text{OH}^-$  ions according to (e.g. [61,77])



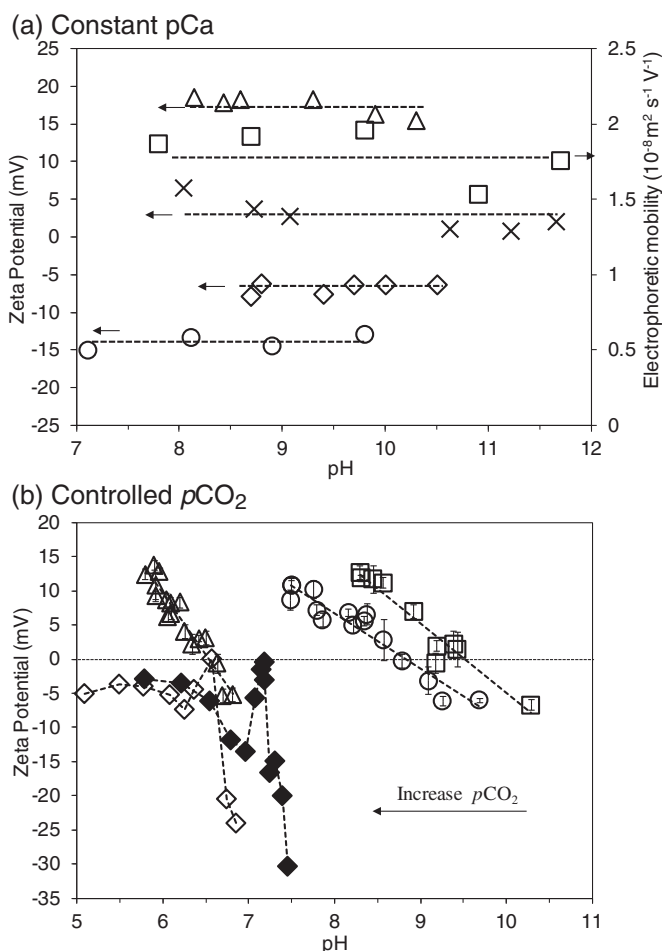
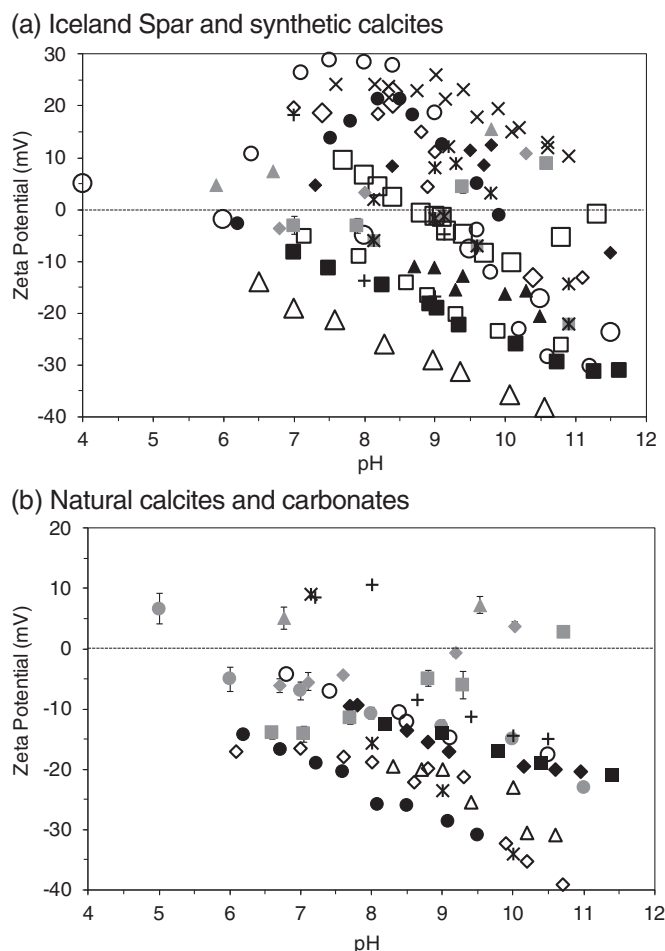
The relative abundance of negatively and positively charged surface sites depends on the relative concentration of  $\text{H}^+$  and  $\text{OH}^-$  in solution; these ions behave as 'potential determining ions' (PDIs) for the surface. Consequently, the net surface charge depends on the solution pH. It is therefore common to see surface charge and/or zeta potential plotted as a function of pH, with the surface charge and/or zeta potential becoming more negative as the pH increases and the deprotonation reaction (4) is favoured, and more positive as the pH decreases and the protonation reaction (5) is favoured (e.g. [13,32]). When the number of positively and negatively charged surface sites exactly balances, a surface charge of zero is observed and the corresponding pH is termed the 'point of zero charge' (PZC). A zeta potential of zero may also be observed and the corresponding pH is termed the 'isoelectric-point' (IEP) (e.g. [34]). The IEP and PZC may not coincide, for reasons discussed in more detail in the next section.

The surface of calcite differs from the metal oxides in several important respects. First, the mineral is soluble in aqueous solution and the lattice ions  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  can be released into solution or deposited on the surface depending upon the solution pH [66,75]. Second, atmospheric  $\text{CO}_2$  in open system experiments can dissolve into solution, affecting the pH and the equilibrium concentrations of  $\text{Ca}^{2+}$ ,  $\text{CO}_3^{2-}$  and  $\text{HCO}_3^-$  (e.g. [18]). Third, the hydrated calcite surface contains protonated anion surface sites  $>\text{CO}_3\text{H}^0$  as well as hydroxylated cation sites  $>\text{CaOH}^0$  (e.g. [57,79]). Evidence to support the presence of these sites has been provided by X-ray photoelectron spectroscopic (XPS) and infrared (IR) spectroscopic measurements [56,72,73]. The protonation and deprotonation reactions at these surface sites, and therefore the relative abundance of positively and negatively charged sites on the mineral surface, is pH dependent according to the following reactions (e.g. [29, 57,79])



Thus it would be expected that the surface charge, and hence the zeta potential, of calcite in aqueous solution are both determined by the pH, with  $\text{H}^+$  and  $\text{OH}^-$  behaving as the PDIs. Numerous studies have determined the variation in zeta potential with pH for a variety of calcite types and solution compositions (e.g. Fig. 2).

However, this analysis ignores the contribution of the lattice ions  $\text{Ca}^{2+}$ ,  $\text{CO}_3^{2-}$  in solution. Several experimental studies have shown that



**Fig. 2.** Zeta potential of (a) Iceland spar and synthetic calcite and (b) natural calcite and carbonate rocks in indifferent electrolytes as a function of pH. Measurements were obtained using the electrophoretic mobility method (EPM), with the exception of Somasundaran and Agar [67] (diamonds) and Thompson and Pownall [78] (squares). Error bars are shown when reported by the source. In (a), data from Somasundaran and Agar [67] in deionized water after different mixing times (♦: no mixing; ◇: after 1 week; ○: after two months of mixing); Sampat Kumar et al. [60] in deionized water (+); Siffert and Fimbel [63] in  $10^{-2}$  M NaCl electrolyte (\*: mass dispersed = 50 mg; grey \*: 30 mg); Thompson and Pownall [78] in various electrolytes (■: NaCl ( $5 \times 10^{-3}$  M); □: NaCl ( $5 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $1 \times 10^{-3}$  M); □: NaCl ( $5 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $1 \times 10^{-3}$  M)/H<sub>2</sub>CO<sub>3</sub>/Ca(OH)<sub>2</sub>); Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte (▲); Vdović and Biščan [81] in  $10^{-3}$  M NaCl electrolyte (○); Vdović [80] in  $10^{-3}$  M NaCl electrolyte (●); Eriksson et al. [18] in  $5.56 \times 10^{-3}$  M NaCl electrolyte (×); Sondi et al. [70] in  $10^{-3}$  M NaCl electrolyte (Δ) and Kasha et al. [42] in deionized water (○). Grey symbols show data from Mahani et al. [47] in various electrolytes (▲: formation brine; ♦: seawater; ■: 25 times diluted seawater). In (b), data from Mishra [48] in  $2 \times 10^{-2}$  M NaClO<sub>4</sub> electrolyte (+); Rao et al. [58] in  $2 \times 10^{-3}$  M NaClO<sub>4</sub> electrolyte (♦); Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte (Δ); Vdović and Biščan [81] in  $10^{-3}$  M NaCl electrolyte (◇); [53] in  $2 \times 10^{-3}$  M NaNO<sub>3</sub> electrolyte (\*); Vdović [80] in  $10^{-3}$  M NaCl electrolyte (○); limestone (●); lake sediment (●) and Somasundaran et al. [69] in  $10^{-3}$  M KNO<sub>3</sub> electrolyte (■). Grey symbols show data on natural carbonate rocks in various electrolytes (●: deionized water containing 0.09 M CaCl<sub>2</sub> from [11]; ▲: formation brine; ♦: seawater; ■: 25 times diluted seawater, all from [47]).

**Fig. 3.** Zeta potential and electrophoretic mobility of synthetic calcite and Iceland spar as a function of pH in experiments with (a) constant pCa and (b) controlled pCO<sub>2</sub>. Error bars are shown when reported by the source. In (a), all data are from synthetic calcite. Data from Foxall et al. [21] in  $10^{-2}$  M NaCl electrolyte with pCa = 2.1 (□); Thompson and Pownall [78] in  $5 \times 10^{-4}$  M CaCl<sub>2</sub> electrolyte with pCa = 3.3 (×); Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte with pCa = 2 (Δ) and pCa = 3 (◇); and Sondi et al. [70] in  $10^{-3}$  M NaCl electrolyte with pCa = 2 (○). In (b), all data are from powdered Iceland spar in deionized water with various pCO<sub>2</sub>. Data from Moulin and Roques [51] (♦: pCO<sub>2</sub> = 0.023; ◇: pCO<sub>2</sub> = 0.234) and Heberling et al. [29] (Δ: pCO<sub>2</sub> = 1; ○ pCO<sub>2</sub> =  $10^{-3.44}$ ; □: pCO<sub>2</sub> =  $10^{-5.2}$ ).

oxides (e.g. [71]). Instead, the lattice ions Ca<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup> adsorb onto the mineral surfaces through surface complexation reactions such as (e.g. [29,71,79,85])



and, in the presence of CO<sub>2</sub>, reactions such as



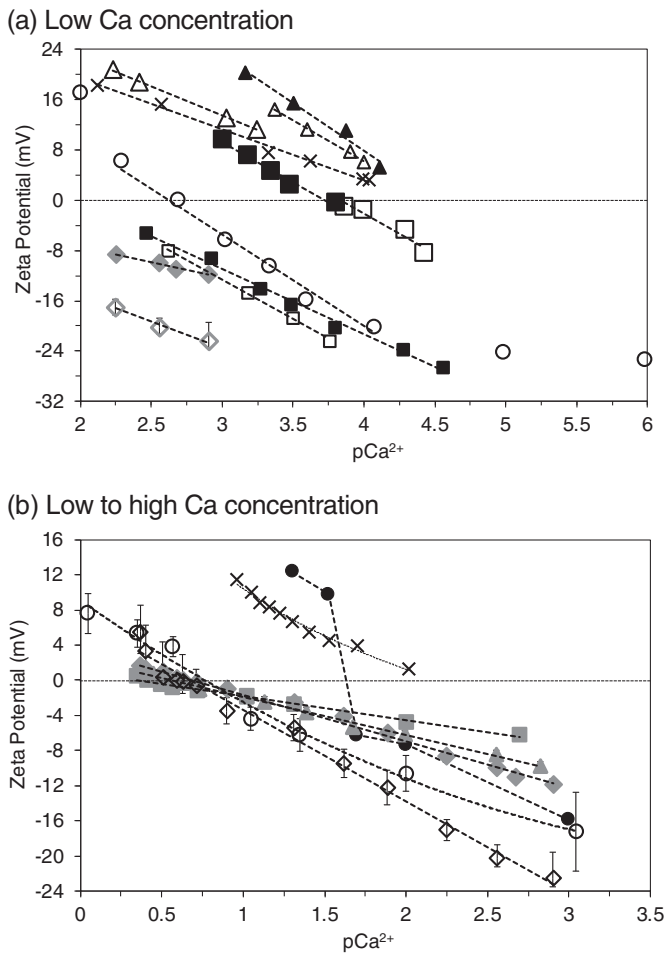
As discussed below, the concentration dependent adsorption of these lattice ions (and other potential determining ions) is the primary control on the zeta potential of calcite.

### 3.2. Potential determining ions

Surface diffraction studies have revealed well-ordered water layers a few Å (=  $1 \times 10^{-10}$  m) above the calcium and carbonate ions on the mineral surface [19,20,24,29,45]. This is the hydrolysis

the zeta potential of calcite is independent of pH if the concentration of calcium (expressed in this paper as pCa, where p represents the negative logarithm) is kept constant (e.g. [12,21,70]; see Fig. 3a). Moreover, other studies have demonstrated a relationship between zeta potential and pCa (e.g. [4,12,54,78]; see Fig. 4), and a strong dependence between the zeta potential of calcite and the excess of Ca<sup>2+</sup> ions at the mineral surface has been observed [33]. These data show that the development of charge on the calcite surface is more complex than the simple protonation and deprotonation reactions that occur at the surface sites of metal





**Fig. 4.** Zeta potential of synthetic calcite and natural carbonate in indifferent electrolytes as a function of pCa, at (a) low Ca concentration only and (b) low to high Ca concentration. Error bars are shown when reported by the source. In (a), all data are from powdered synthetic calcite with the exception of Alroudhani et al. [4] (grey symbols). Data from Thompson and Pownall [78] in different aqueous compositions (■: NaCl ( $2 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $10^{-2}$  M); □: NaCl ( $2 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $10^{-2}$  M)/Ca(OH)<sub>2</sub> ( $10^{-2}$  M); ■: NaCl ( $2 \times 10^{-3}$  M)/NaHCO<sub>3</sub> ( $10^{-2}$  M)/H<sub>2</sub>CO<sub>3</sub> ( $10^{-2}$  M)); Foxall et al. [21] in three different NaCl concentrations (▲:  $1 \times 10^{-2}$  M; △:  $5 \times 10^{-2}$  M; big △:  $1.5 \times 10^{-1}$  M); Huang et al. [33] in 0.01 CaCl<sub>2</sub> electrolyte (×) and Cicerone et al. [12] in  $10^{-3}$  M KCl electrolyte (○). Grey diamonds show data from Alroudhani et al. [4] on natural carbonate in 0.05 M NaCl electrolyte at high pCa (♦: SPM data; ◇: EPM data); the full span of these data is shown in Fig. 2b. In (b), all data are from natural calcite and carbonate rocks, with the exception of Kasha et al. [42] (filled symbols). Data from Zhang and Austad [88] on powdered Stevns Klint chalk in 0.571 M NaCl electrolyte (×); Chen et al. [11] in deionized water (○); Kasha et al. [42] in synthesized brines equivalent to seawater composition (●) and Alroudhani et al. [4] in 0.05 M NaCl (◇). Grey symbols show data from Alroudhani et al. [4] in intact natural carbonate rocks in various NaCl concentrations (♦: 0.05 M; ▲: 0.5 M; ■: 2 M; all from; error bars are the same size or smaller than the symbols).

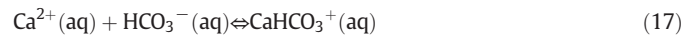
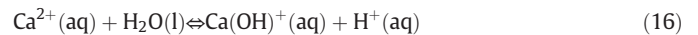
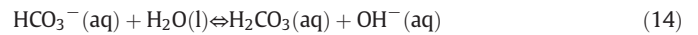
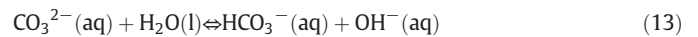
plane of Stipp [71] and the 0-plane invoked in various surface complexation models (SCMs) of the calcite–water interface ([29, 85]; see also Fig. 1a). Protonation and deprotonation reactions occur in this plane, but it does not contain the lattice ions Ca<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup> or other adsorbed ion complexes. These are confined to the 1- and 2-planes (or the a- and b-planes), which here we associate with the inner and outer Helmholtz planes respectively (Fig. 1a). Most SCMs include ordered water layers at the 0-plane, and the pH-dependent protonation and deprotonation reactions (6)–(8) that occur at the sites defining the 0-plane [29,71,85]. However, the SCM of Heberling et al. [29] predicted that the charge at the 0-plane is only weakly pH dependent. The dominant surface species are >CaOH<sup>0</sup> and >CaCO<sub>3</sub><sup>-</sup>, causing the 0-plane to be negatively

charged across the pH range 5.5–11. Thus, unlike the >MeOH groups on oxide mineral surfaces, the calcite surface species >CaOH<sup>-</sup> and >CaCO<sub>3</sub><sup>-</sup> do not determine the pH dependence of the zeta potential. Rather, the potential at the OHP and, hence, the zeta potential, is instead controlled by adsorption of the potential-determining lattice ions Ca<sup>2+</sup> or CO<sub>3</sub><sup>2-</sup> at the 1- or 2-planes. The available evidence suggests that most of the adsorbed lattice PDIs are located at the 2-plane [29,85].

Dissolution of calcite in aqueous solution is described by the following reaction:



The equilibrium conditions are defined by the pH, the concentration of the ionic species and, in open system conditions, the partial pressure of CO<sub>2</sub> (pCO<sub>2</sub>) in solution [66,68]. Moreover, the concentration of lattice ion species available to adsorb onto the calcite surface is controlled by the reactions [18]:



It can be seen from (12)–(17) that pH (and pCO<sub>2</sub>) determines the Ca<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup> concentrations in the solution at equilibrium. Apparent trends between zeta potential and pH therefore reflect the fact that pH and pCa are directly related at fixed pCO<sub>2</sub> (e.g. [64]). We confirm this experimentally in a later section. Thus, H<sup>+</sup> is not the key PDI for the calcite surface; rather, it is the lattice ion Ca<sup>2+</sup> [4,12,21,33,51,54]. Moreover, the PZC and IEP are not closely related. In principle, the PZC could be defined in terms of pH, as pH controls the protonation and deprotonation reactions at the 0-plane. However, in practice, the PZC is not observed over the pH range 5.5–11 [29]. The IEP must be defined in terms of pCa, or in terms of the concentration of other PDIs for the calcite surface such as Mg<sup>2+</sup> as discussed below.

#### 4. Experimental measurements of zeta potential

Published measurements of zeta potential in calcite and natural carbonates in aqueous solution range from +29 mV to −39 mV (Fig. 2). As discussed below, the zeta potential depends on the electrolyte concentration and composition, the nature of the calcite, sample preparation, experimental conditions and measurement method. However, limited reporting of the nature of the samples and experimental method, especially sample history, preparation and cleaning, and experimental conditions (especially pH, pCa and pCO<sub>2</sub>), makes it challenging to systematically compare experimental data (see Table 1 for a summary). In this section, we review and discuss the data published to date, and summarize the key controls on zeta potential.

##### 4.1. Electrolyte concentration and composition

###### 4.1.1. Effect of pH

Many studies have reported more negative values of zeta potential with increasing pH (e.g. [12,18,54,78,80]). For example, Thompson and Pownall [78] reported a close-to-linear relationship between zeta potential and pH (square data in Fig. 2a) using packed synthetic calcite and NaCl solution of ionic strength  $5 \times 10^{-3}$  M

**Table 1**

Summary of published zeta potential of calcite and carbonate. Table 1a reports studies focusing on the effect of varying pH. Table 1b reports studies focusing on the effect of varying pCa.

| 1a                           |        |                                                                                                                                |                               |                         |                                    |                                        |                       |                   |                   |           |
|------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------|------------------------------------|----------------------------------------|-----------------------|-------------------|-------------------|-----------|
| Material                     | Method | Background electrolyte                                                                                                         | Equilibrium condition         | Equilibration time (hr) | Equilibrium pH (pH <sub>EQ</sub> ) | Open vs. closed system                 | Zeta potential at EQ. | pH <sub>IEP</sub> | IEP determination | Reference |
| Iceland spar                 | SPM    | Deionized water (HNO <sub>3</sub> /NaOH)                                                                                       | Calcite/water/air             | 1440                    | 8.2                                | Open                                   | Positive              | 8.2               | Extrapolated      | [67]      |
| Iceland spar                 | SPM    | SiO <sub>2</sub> /Na <sub>2</sub> O                                                                                            | Calcite/water                 | N/A                     | 9.8                                | Closed                                 | Positive              | 10.8              | Direct            | [22]      |
| Iceland spar                 | SPM    | Deionized water                                                                                                                | Calcite/water/CO <sub>2</sub> | N/A                     | 6.55                               | pCO <sub>2</sub> = 0.23                | Negative              | 6.55              | Direct            | [51]      |
|                              |        |                                                                                                                                |                               |                         | 7.05                               | pCO <sub>2</sub> = 0.023               | Negative              | 7.05              |                   |           |
| Iceland spar                 | SPM    | Deionized water                                                                                                                | Calcite/water/CO <sub>2</sub> | 24–720                  | 5.8                                | pCO <sub>2</sub> = 1                   | Negative              | 6.5               | Direct            | [29]      |
|                              |        |                                                                                                                                |                               |                         | 7.5                                | pCO <sub>2</sub> = 10 <sup>-3.44</sup> | Negative              | 8.8               |                   |           |
|                              |        |                                                                                                                                |                               |                         | 8.3                                | pCO <sub>2</sub> = 10 <sup>-5.2</sup>  | Negative              | 9.4               |                   |           |
| Iceland spar                 | EPM    | Deionized water                                                                                                                | Calcite/water                 | 24                      | N/A                                | Closed                                 | Negative              | 5.4               | Extrapolated      | [42]      |
| Iceland spar                 | EPM    | Deionized water                                                                                                                | Calcite/water/CO <sub>2</sub> | 1–2                     | N/A                                | Open                                   | Negative              | 7.2–7.8           | Extrapolated      | [60]      |
| Iceland spar                 | EPM    | Formation water                                                                                                                | Calcite/water                 | 24                      | N/A                                | Closed                                 | Positive              | N/A               | Direct            | [47]      |
|                              |        | Seawater (SW)                                                                                                                  |                               |                         | N/A                                |                                        | Negative              | 7.5               |                   |           |
|                              |        | 25 times diluted SW                                                                                                            |                               |                         | 9.2                                |                                        | Negative              | 8.5               |                   |           |
| Synthetic calcite            | EPM    | 1 × 10 <sup>-2</sup> ,<br>5 × 10 <sup>-2</sup> ,<br>1.5 × 10 <sup>-1</sup> M NaCl                                              | Calcite/water/air             | N/A                     | N/A                                | Open                                   | Positive              | N/A               | N/A               | [21]      |
| Synthetic calcite            | EPM    | 10 <sup>-2</sup> M NaCl                                                                                                        | Calcite/water                 | 10                      | 9.1                                | Closed                                 | Pos – Neg             | 10–11             | N/A               | [63]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-3</sup> M NaCl                                                                                                    | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Negative              | N/A               | N/A               | [78]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-3</sup> M NaCl/<br>10 <sup>-3</sup> M NaHCO <sub>3</sub>                                                          | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Negative              | N/A               | N/A               | [78]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-3</sup> M NaCl/<br>10 <sup>-3</sup> M NaHCO <sub>3</sub> /<br>H <sub>2</sub> CO <sub>3</sub> /Ca(OH) <sub>2</sub> | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Negative              | 8.9               | Extrapolated      | [78]      |
| Synthetic calcite            | SPM    | 5 × 10 <sup>-4</sup> M CaCl <sub>2</sub>                                                                                       | Calcite/water                 | 48                      | 9.11–9.87                          | Closed                                 | Positive              | N/A               | N/A               | [78]      |
| Synthetic calcite            | EPM    | 10 <sup>-2</sup> M NaCl                                                                                                        | Calcite/water                 | 24                      | 10                                 | Closed                                 | Positive              | 9.6               | Direct            | [54]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M KCl                                                                                                         | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Negative              | N/A               | N/A               | [12]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M KCl/10 <sup>-3</sup> M<br>CaCl <sub>2</sub>                                                                 | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Negative              | N/A               | N/A               | [12]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M KCl/10 <sup>-2</sup> M<br>CaCl <sub>2</sub>                                                                 | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Positive              | N/A               | N/A               | [12]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.6                                | Closed                                 | Positive              | 9.8               | Direct            | [81]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.4                                | Closed                                 | Positive              | 9.8               | Direct            | [80]      |
| Synthetic calcite            | EPM    | 5.56 × 10 <sup>-3</sup> M NaCl                                                                                                 | Calcite/water/CO <sub>2</sub> | 1                       | 8–8.5                              | Open                                   | Positive              | N/A               | N/A               | [18]      |
| Synthetic calcite            | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.3                                | Closed                                 | Negative              | N/A               | N/A               | [70]      |
| Natural calcite              | EPM    | 2 × 10 <sup>-3</sup> M NaClO <sub>4</sub>                                                                                      | Calcite/water                 | 0.5                     | 9–10                               | Closed                                 | Negative              | N/A               | N/A               | [58]      |
| Natural calcite              | EPM    | 2 × 10 <sup>-2</sup> M NaClO <sub>4</sub>                                                                                      | Calcite/water/air             | 0.5                     | 8.2                                | Open                                   | Negative              | 8.2               | Extrapolated      | [48]      |
| Natural calcite              | EPM    | 10 <sup>-2</sup> M NaCl                                                                                                        | Calcite/water                 | 24                      | 10                                 | Closed                                 | Negative              | 9.4               | Direct            | [54]      |
| Natural (biogenetic) calcite | EPM    | 10 <sup>-3</sup> M KCl                                                                                                         | Calcite/water                 | 168                     | 9.8–10                             | Closed                                 | Negative              | N/A               | N/A               | [12]      |
| Natural calcite              | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.6                                | Closed                                 | Negative              | N/A               | N/A               | [81]      |
| Natural calcite              | EPM    | 2 × 10 <sup>-3</sup> M NaNO <sub>3</sub>                                                                                       | Calcite/water/air             | 2.2                     | 8.3                                | Open                                   | Negative              | 7.8               | Extrapolated      | [53]      |
| Natural calcite              | EPM    | 10 <sup>-3</sup> M NaCl                                                                                                        | Calcite/water/air             | Few                     | 8.4                                | Closed                                 | Negative              | N/A               | N/A               | [80]      |
| Natural calcite              | EPM    | 10 <sup>-3</sup> M KNO <sub>3</sub>                                                                                            | Calcite/water/air             | 1                       | N/A                                | Open                                   | Negative              | N/A               | N/A               | [69]      |
| Natural carbonate            | EPM    | Deionized water                                                                                                                | Calcite/water                 | 1                       | N/A                                | Closed                                 | Negative              | 5.2–5.8           | Extrapolated      | [11]      |
| Natural carbonate            | EPM    | Formation water                                                                                                                | Calcite/water                 | 24                      | N/A                                | Closed                                 | Positive              | N/A               | Direct            | [47]      |
|                              |        | Seawater (SW)                                                                                                                  |                               |                         | N/A                                |                                        | Negative              | 9.25              |                   |           |
|                              |        | 25 times diluted SW                                                                                                            |                               |                         | 9.2                                |                                        | Negative              | 10.25             |                   |           |

| 1b                |                         |                                                                                                                          |                       |                         |                                    |                        |                        |                    |                   |           |
|-------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------|------------------------------------|------------------------|------------------------|--------------------|-------------------|-----------|
| Material type     | Method                  | Background electrolyte                                                                                                   | Equilibrium condition | Equilibration time (hr) | Equilibrium pH (pH <sub>EQ</sub> ) | Open vs. closed system | dζ/dpCa                | pCa <sub>IEP</sub> | IEP determination | Reference |
| Iceland spar      | Electro osmosis         | NaCl/Na <sub>2</sub> CO <sub>3</sub> /CaCl <sub>2</sub>                                                                  | Calcite/water         | 2                       | 9                                  | Closed                 | N/A                    | 3.33               | Extrapolated      | [16]      |
| Iceland spar      | SPM on packed particles | Deionized water (HNO <sub>3</sub> /NaOH)                                                                                 | Calcite/water/air     | 1440                    | 8.2                                | Open                   | N/A                    | 4.25               | Extrapolated      | [67]      |
| Iceland spar      | EPM                     | 0.571 M NaCl                                                                                                             | Calcite/water         | 24                      | N/A                                | Closed                 | N/A                    | 1.65–1.75          | Extrapolated      | [42]      |
| Synthetic calcite | EPM                     | 1 × 10 <sup>-2</sup> ,<br>5 × 10 <sup>-2</sup> ,<br>1.5 × 10 <sup>-1</sup> M NaCl                                        | Calcite/water/air     | N/A                     | N/A                                | Open                   | –15.2<br>–12.8<br>–9.3 | 4.5                | Extrapolated      | [21]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> M NaCl                                                                                              | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –12.7                  | 2.02               | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> M<br>(NaCl/NaHCO <sub>3</sub> )                                                   | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –10.2                  | 1.92               | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> M<br>(NaCl/CaCl <sub>2</sub> )                                                    | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –10.1                  | 3.4                | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> M<br>(NaCl/H <sub>2</sub> CO <sub>3</sub> )                                       | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –10.8                  | 4                  | Extrapolated      | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> /10 <sup>-2</sup> M<br>(NaCl/NaHCO <sub>3</sub> /H <sub>2</sub> CO <sub>3</sub> ) | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –12.9                  | 3.8                | Direct            | [78]      |
| Synthetic calcite | SPM                     | 2 × 10 <sup>-3</sup> /10 <sup>-2</sup> /10 <sup>-2</sup> M                                                               | Calcite/water         | 48                      | 9.11–9.87                          | Closed                 | –12.2                  | 3.8                | Direct            | [78]      |

Table 1 (continued)

| Material type       | Method                | Background electrolyte                                  | Equilibrium condition | Equilibration time (hr) | Equilibrium pH (pH <sub>EQ</sub> ) | Open vs. closed system | dζ/dpCa                  | pCa <sub>IEP</sub> | IEP determination | Reference |
|---------------------|-----------------------|---------------------------------------------------------|-----------------------|-------------------------|------------------------------------|------------------------|--------------------------|--------------------|-------------------|-----------|
| 1b                  |                       |                                                         |                       |                         |                                    |                        |                          |                    |                   |           |
| Synthetic calcite   | EPM                   | (NaCl/NaHCO <sub>3</sub> /Ca(OH) <sub>2</sub> )         | Calcite/water         | 24                      | 10                                 | Closed                 | N/A                      | 3.3                | Direct            | [54]      |
| Synthetic calcite   | SPM                   | 10 <sup>−2</sup> M NaCl                                 | Calcite/water         | 120                     | 9.3–9.9                            | Closed                 | −8.34                    | 4.38               | Extrapolated      | [33]      |
| Synthetic calcite   | EPM                   | 10 <sup>−5</sup> –0.01 M CaCl <sub>2</sub>              | Calcite/water         | 168                     | 9.8–10                             | Closed                 | −14.5                    | 2.7                | Direct            | [12]      |
|                     |                       | 10 <sup>−3</sup> M KCl                                  | Calcite/water         |                         |                                    |                        | (in the vicinity of IEP) |                    |                   |           |
| Natural calcite     | EPM                   | 10 <sup>−2</sup> M NaCl                                 | Calcite/water         | 24                      | 10                                 | Closed                 | N/A                      | 4                  | Direct            | [54]      |
| Stevens Klint chalk | EPM                   | CaCl <sub>2</sub> and NaSO <sub>4</sub> in 0.571 M NaCl | Chalk/water/air       | 48                      | Controlled at 8.4                  | Closed                 | N/A                      | N/A                | N/A               | [88]      |
| Natural carbonate   | EPM                   | Deionized water (DI)                                    | Calcite/water         | 1                       | Controlled at 8.4                  | Closed                 | N/A                      | 0.2–0.75           | Extrapolated      | [11]      |
| Natural carbonate   | SPM in intact samples | 0.05 M NaCl                                             | Calcite/water/air     | 168–720                 | 8.2                                | Closed                 | −5.29                    | 0.61               | Direct            | [4]       |
|                     |                       | 0.5 M NaCl                                              |                       |                         |                                    |                        | −4.35                    | 0.49               |                   |           |
|                     |                       | 2 M NaCl                                                |                       |                         |                                    |                        | −2.75                    | 0.41               |                   |           |

(filled squares) and NaCl/NaHCO<sub>3</sub> of ionic strength  $5 \times 10^{-4}$  M (open squares). Similarly, Vdovic [80] obtained a linear relationship between zeta potential and pH using natural carbonate powder samples suspended in  $10^{-3}$  M NaCl electrolyte (filled and open circles in Fig. 2b). However, there are numerous exceptions (e.g. [11,42,46–48,63,67,81]). For example, Siffert and Fimbel [63] found that the zeta potential of synthetic calcite at equilibrium pH (9.1; Table 1a) could be either positive or negative, dependent on the dispersed mass. Away from the equilibrium pH, the zeta potential decreased with either increasing or decreasing pH (open and grey stars in Fig. 2a). Vdović and Bišćan [81] (open circles in Fig. 2a) and Vdovic [80] (solid circles in Fig. 2a) observed similar behaviour. The latter reported zeta potential values that changed polarity with both increasing and decreasing pH, thus apparently defining two IEPs. Mahani et al. [46] [47] reported an increasingly negative zeta potential with increasing pH (grey triangles, diamonds and squares in Fig. 2a–b), contrary to the trends reported above. However, as discussed in the previous section, SCMs for calcite suggest the electrical potential at the mineral surface is only weakly pH dependent, and experimental data have shown that the zeta potential is independent of pH if pCa is held constant (Fig. 3). These data confirm that zeta potential is not directly controlled by pH, but rather by the concentration of lattice ions such as Ca<sup>2+</sup> and other PDI's discussed below.

#### 4.1.2. Effect of pCa and pCO<sub>2</sub>

Many authors have concluded that the principal PDI's for calcite are the lattice ions Ca<sup>2+</sup> and CO<sub>3</sub><sup>2−</sup> (e.g. [7,12,21,33,48,54,62,63,67,78,80,81]). Consequently, the effect of Ca<sup>2+</sup> on the zeta potential has been the topic of several experimental studies (e.g. [4,12,33,54,78]; see Fig. 4). The consistent observation in these studies is that decreasing pCa yields more positive zeta potential, consistent with increased adsorption of Ca<sup>2+</sup> onto the calcite surface (Fig. 1). However, significant differences in the magnitude and polarity of the zeta potential, in the gradient of zeta potential vs. pCa (dζ/dpCa), and in the IEP (expressed as pCa), have been reported (Fig. 4; Table 1b). The zeta potential varies between +20 mV and −26 mV, depending on calcite type and experimental conditions. Most studies report a linear or close-to-linear relationship between zeta potential and pCa, especially for pCa values close to the IEP, with the gradient dζ/dpCa in the range −8 to −15 mV/decade (Fig. 4). A linear relationship is consistent with Nernstian behaviour of the calcite surface and suggests the electrical double layer can be

reasonably described by the simple Gouy–Chapman–Grahame model close to the IEP (e.g. [34]). This model is valid only when ions other than the lattice ions show no specific adsorption at the surface. The relationship between zeta potential and pCa can then be expressed as [21]:

$$\left. \frac{d\zeta}{dpCa} \right|_{\zeta \rightarrow 0} = \frac{-2.303 \frac{kT}{ze}}{\left(1 + \frac{C_d}{C_s}\right) \exp(\kappa\Delta)}, \quad (18)$$

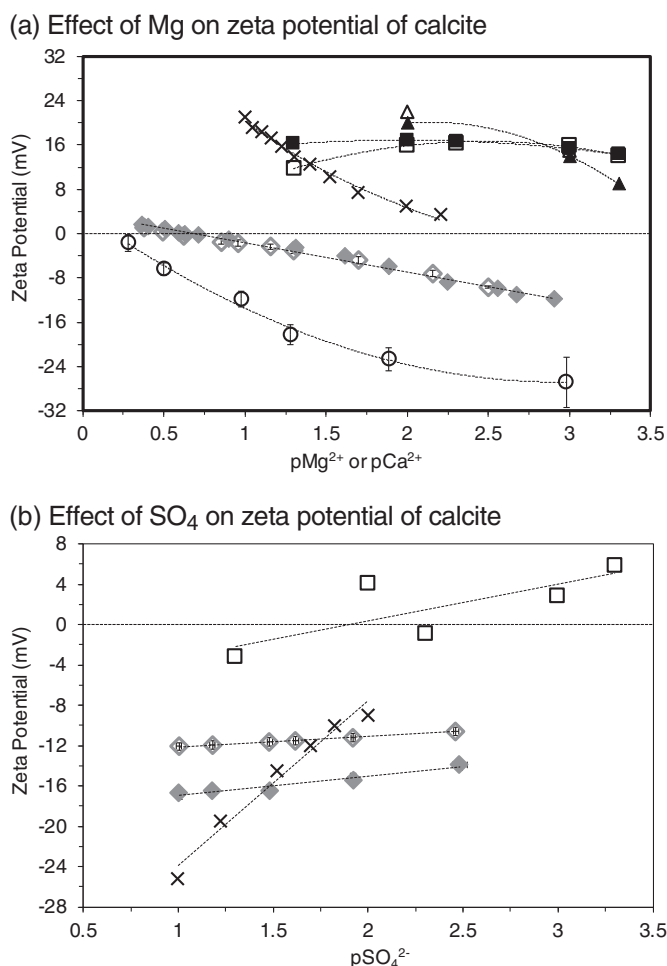
where  $k$  is Boltzmann's constant,  $T$  is the temperature,  $z$  is the valence of the PDI,  $e$  is the charge on an electron,  $\kappa$  is the Debye–Hückel reciprocal length,  $C_d$  and  $C_s$  are the capacitance per unit area of the diffuse and Stern layers respectively,  $\kappa$  is the Debye–Hückel reciprocal length, and  $\Delta$  is the distance of the shear plane from the Stern plane (see Fig. 1).

Zeta potential data obtained at low ionic strength using the EPM method show a broadly consistent value of dζ/dpCa of approximately −13 mV/decade (Fig. 4a). Eq. (18) can be used to match these data using one value of  $C_s \sim 0.4 \text{ Fm}^{-2}$  and  $\Delta = \text{zero}$ . For comparison, Heberling et al. [29] fitted their zeta potential data using a more sophisticated SCM with  $C_s = 0.45 \text{ Fm}^{-2}$  and  $\Delta = 0.33 \text{ nm}$ . Previous SCMs for calcite used unrealistically high  $C_s$  values of 10–100  $\text{Fm}^{-2}$  [57,79]. However, a number of recent studies have reported non-linear relationships between zeta potential and pCa (e.g. [11,42,88]; see Fig. 4b). Moreover, a wide range of IEP (expressed as pCa) values have been reported (pCa of 1.92–4.5; Table 1b). As discussed in the following sections, the variability of these results might be attributed to differences in experimental conditions, including pCO<sub>2</sub>, calcite type, measurement technique and the establishment of equilibrium. The effect of the carbonate ion on the zeta potential has received less attention than the calcium ion. The very limited data suggest that addition of CO<sub>3</sub><sup>2−</sup> makes the zeta potential more negative, consistent with increased adsorption of CO<sub>3</sub><sup>2−</sup> onto the calcite surface (e.g. [16,51,64]).

#### 4.1.3. Effect of pMg and pSO<sub>4</sub>

The magnesium ion is also compatible with the calcite crystal structure [8]. It has a direct influence on geochemical processes involving natural carbonates of mixed mineralogy such as dolomite [50] and is also abundant in natural brines, yet has received much less attention than Ca<sup>2+</sup>. De Groot and Duyvis [14] suggested that Ca<sup>2+</sup> and Mg<sup>2+</sup> influence the zeta potential of calcite to a similar extent, but their dataset





**Fig. 5.** Zeta potential of synthetic calcite and natural carbonate as a function of (a) pMg and (b) pSO<sub>4</sub>. Error bars are shown when reported by the source. In (a), open symbols show zeta potential as a function of pMg. Data from De Groot and Duyvis [14] in deionized water (Δ); Smallwood [64] in deionized water (□); Zhang et al. [87] in 0.571 M NaCl electrolyte (×); Chen et al. [11] in deionized water (○) and Alroudhani et al. [4] in intact natural Portland carbonate in 0.05 M NaCl electrolyte (◇; error bars for these data are the same size or smaller than the symbols). For comparison, filled symbols show measurements as a function of pCa when available from the same studies. In (b), data from Smallwood [64] on synthetic calcite in deionized water (□); Zhang and Austad [88] on Stevns Klint chalk in 0.571 M NaCl electrolyte (×) and Alroudhani et al. [4] in intact natural Portland carbonate in various NaCl electrolytes (◆: 0.05 M; ◇: 0.5 M; error bars are the same size or smaller than the symbols; data from).

was very sparse, comprising just five measurements (two for Mg<sup>2+</sup> and three for Ca<sup>2+</sup>) at high pMg (i.e. at very low concentration; compare filled and open triangles in Fig. 5a). Smallwood [64] obtained a similar response (compare filled and open squares in Fig. 5a) over the same concentration range. More recent studies have found that the zeta potential becomes more positive with decreasing pMg ([4,11,87]; see data represented by crosses, circles and diamonds in Fig. 5a). Alroudhani et al. [4] found that Mg<sup>2+</sup> behaved identically to Ca<sup>2+</sup> within experimental error over a wide range of pCa and pMg (0.4–3; compare filled and open diamonds). Moreover, they were the first to report an IEP in terms of pMg, which was identical to pCa<sub>IEP</sub> within experimental error. However, data showing the effect of pMg on zeta potential are still very scarce, especially at low pMg.

A number of studies have also shown that the zeta potential is affected by the concentration of SO<sub>4</sub><sup>2-</sup>, observing that the zeta potential becomes more negative with decreasing pSO<sub>4</sub> ([4,64,88]; Fig. 5b). Smallwood [64] argued that SO<sub>4</sub><sup>2-</sup> is adsorbed onto the calcite mineral

surface, similar to the lattice ions Ca<sup>2+</sup>, Mg<sup>2+</sup> and CO<sub>3</sub><sup>2-</sup>, as indicated by its ability to reverse the polarity of the zeta potential (Fig. 5b). However, no other papers have reported an IEP in terms of pSO<sub>4</sub>. Zhang and Austad [88] and Alroudhani et al. [4] both observed a linear relationship between zeta potential and pSO<sub>4</sub> for chalk and natural carbonate respectively, but the IEP was not encountered over the concentration range investigated (Fig. 5b). Data showing the effect of pSO<sub>4</sub> on zeta potential are still scarce and do not conclusively show that SO<sub>4</sub><sup>2-</sup> is a PDI for the calcite surface.

#### 4.1.4. Indifferent ions and total ionic strength

Ions other than Ca<sup>2+</sup>, Mg<sup>2+</sup>, CO<sub>3</sub><sup>2-</sup> and SO<sub>4</sub><sup>2-</sup> are generally assumed to be indifferent to the calcite mineral surface; they have little or no tendency to adsorb onto the surface. It is well known that the zeta potential decreases in magnitude with increasing concentration of indifferent ions such as Na<sup>+</sup>, K<sup>+</sup>, Cl<sup>-</sup>, reflecting contraction of the electrical double layer (EDL; e.g. [36,83]). Pierre et al. [54] obtained the same IEP (expressed as pH) on natural calcite irrespective of the concentration of NaCl (0, 10<sup>-3</sup>, 10<sup>-2</sup> M), indicating that Na and Cl are indifferent ions. Alroudhani et al. [4] found that the sensitivity of zeta potential to pCa (i.e. dζ/dpCa) decreased as NaCl concentration increased, consistent with contraction of the EDL reducing the absolute magnitude of the zeta potential (Fig. 4b and Table 1b). However, they also found that the IEP (as pCa) decreased with increasing NaCl concentration, which is not expected for indifferent ions. They suggested that at the high NaCl concentrations investigated (up to 2 M), the ability of Ca<sup>2+</sup> ions to interact with calcite surface was reduced due to contraction of the EDL and increasing occupancy of hydrated Na<sup>+</sup> ions in the diffuse layer. However, data to support this hypothesis are scarce; few studies have obtained zeta potential measurements on calcite at high ionic strength (>0.1 M).

#### 4.2. Partial pressure of CO<sub>2</sub>

Variations in pCO<sub>2</sub> across different experiments may be one factor that explains the variation in reported values of IEP. In closed system experiments with controlled pCO<sub>2</sub>, the very limited data available suggest that the IEP (expressed as pH) decreases with increasing pCO<sub>2</sub> [29,51]. Moreover, in open system experiments, Somasundaran and Agar [67] postulated that pH<sub>IEP</sub> was a function of mixing time of the sample powder in solution, due to an increase in CO<sub>2</sub> concentration in the suspension. They used this mechanism to explain the decrease in pH<sub>IEP</sub> they observed with increased mixing time. The reported pH<sub>IEP</sub> in closed system experiments is generally higher than the value in open system experiments, consistent with reduced access to CO<sub>2</sub> (Table 1a). The absence of, or limited access to, CO<sub>2</sub> in closed system experiments causes a decrease in carbonic acid formation (Eq. (16)) and in turn increases pH<sub>IEP</sub> [66].

The effect of pCO<sub>2</sub> on the IEP expressed as pCa or pMg is less well understood. No studies have varied pCO<sub>2</sub> and pCa or pMg, or measured pCa or pMg whilst varying pCO<sub>2</sub> and pH. Studies investigating the effect of pCa on zeta potential have typically been conducted in simple open or closed system conditions. In open system experiments, measured pCa<sub>IEP</sub> lies in the range of 4.25–4.5 ([21,33,67]; Table 1b). In closed system experiments, reported pCa<sub>IEP</sub> values have a wider range of 0.4–4 and are always lower than the pCa<sub>IEP</sub> values reported in open system experiments (Table 1b). The broad range of pCa<sub>IEP</sub> observed in closed system experiments might be attributed to different amounts of CO<sub>2</sub> dissolved into solution during preparation (e.g. [78]) or the type of calcite used (synthetic vs. natural; [54]; see also the next section). Recent studies of zeta potential on natural carbonates in closed system experiments consistently show the lowest values of pCa<sub>IEP</sub> in the range 0.4–0.75 [4, 11,42]. In closed system experiments, pCO<sub>2</sub> is fixed and, as we show later, lower pCa corresponds to lower pH. Lower pH is equivalent to higher pCO<sub>2</sub> (e.g. [29]; see also reactions 14–18).

#### 4.3. Natural versus synthetic calcite

The zeta potential has been measured on a wide variety of calcite samples with differing nature and origin which can be broadly subdivided into three types: (1) synthetic calcite (pure precipitated crystalline calcium carbonate), (2) Iceland spar (pure crystalline calcite of natural origin) and (3) natural carbonate (composed mostly of calcite with inorganic material or organic matter). Data for Iceland spar and synthetic calcite show both positive and negative values for a given value of pH (Fig. 2; Table 1a) and pCa (Fig. 4a, Table 1b). The broad range of zeta potential values observed for a given pH may reflect differences in the concentrations of PDIs such as  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  that were not measured in the experiments; they may also reflect differences in  $p\text{CO}_2$  or, as discussed below, the equilibration method. The range of values observed for a given pCa is more difficult to explain, but most likely reflects differences in  $p\text{CO}_2$  or the equilibration method; it may also reflect differences in the concentrations of other PDIs such as  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$ .

Data for natural calcite and carbonate rocks show less variability, with most studies returning negative values of zeta potential irrespective of pH; the few positive values in Fig. 2 were mostly obtained using natural brines rich in  $\text{Ca}^{2+}$  where, as we show later, positive zeta potential is expected. Berlin and Khabakov [7] reported negative zeta potentials on 115 natural carbonate samples irrespective of pH. Likewise, with relatively few exceptions, the zeta potential on natural samples has been found to be negative at high pCa and positive at low pCa, although there is some variability in  $p\text{Ca}_{\text{IEP}}$  as discussed above (Fig. 4b). It has been suggested that the more negative zeta potentials typically observed in natural carbonates are due to the presence of organic matter incorporated in their structure during formation [80,81]. For example, Vdovic [80] obtained negative zeta potential on two natural carbonate samples over the whole pH range (filled and open circles in Fig. 2b); however, they obtained a positive zeta potential on synthetic calcite (filled circles in Fig. 2a). Likewise, Cicerone et al. [12] found that natural (biogenetic) calcite (open triangles in Fig. 2b) bore a more negative zeta potential than pure calcite (filled triangles in Fig. 2a). However, the role of organic material in controlling the zeta potential of natural carbonates remains poorly understood.

#### 4.4. Measurement technique

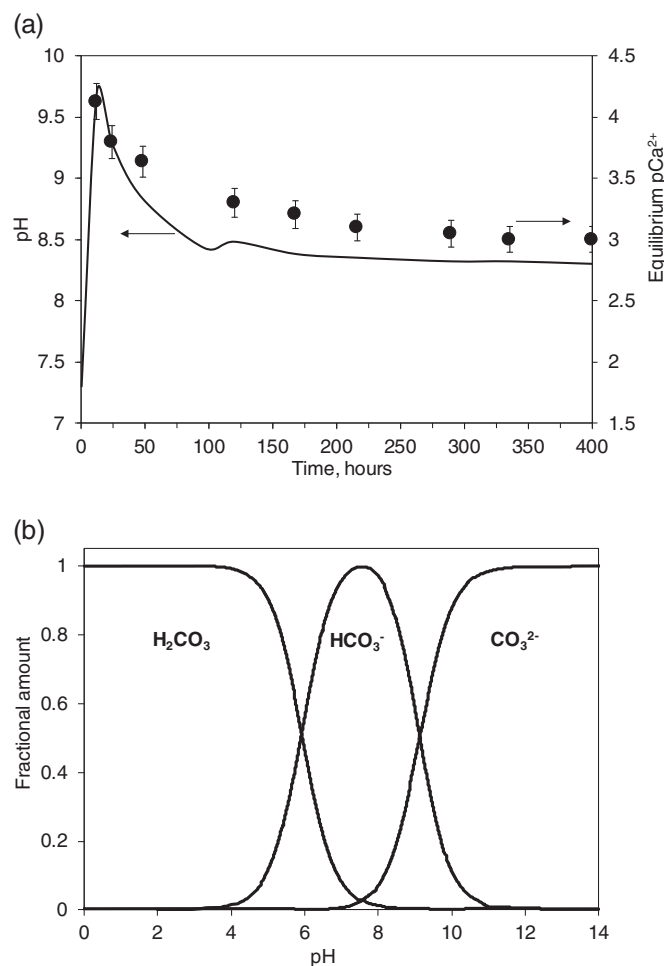
Measurements of zeta potential generally use either the electrophoretic mobility (EPM) method or the streaming potential method (SPM) introduced earlier. There appears to be no systematic difference in the reported IEP and both techniques have yielded a linear Nernstian relationship between zeta potential and pCa (Table 1; Fig. 4). However, only Alroudhani et al. [4] have compared the two techniques on the same (natural) calcite samples and electrolytes. Using the EPM method and natural (powdered) carbonate suspended in 0.05 M NaCl electrolyte, they obtained a linear regression between zeta potential and pCa with a gradient of  $-10.45 \text{ mV/decade}$ , which they fitted using Eq. (18) and values of  $C_s = 1.13 \text{ Fm}^{-2}$  and  $\Delta = 0$ . Note that this gradient is lower than obtained in previous studies on similar samples (as discussed above; see also Fig. 4a) and the value of  $C_s$  used to obtain a match is higher. However, Alroudhani et al. [4] investigated electrolytes of higher total ionic strength than previous studies and their results suggest that  $C_s$  increases with increasing total ionic strength.

Using the SPM method and the same natural (intact) carbonates and electrolytes, Alroudhani et al. [4] obtained the same IEP (expressed as pCa) within experimental error, but the gradient of a linear regression through the data was approximately two times smaller at  $-5.10 \text{ mV/decade}$ . They argued that the shear plane is further away from the mineral surface in intact natural samples as opposed to powdered samples owing to the complex topology of the pore-space (see also Vernhet et al. [82] for similar comparisons on other materials), and fitted the SPM data using Eq. (18) with the same value of  $C_s$  but a higher value of

$\Delta = 0.245 \text{ nm}$ . Thus differences in the observed sensitivity of zeta potential to pCa (or the concentration of other PDIs) may reflect differences in the measurement technique and/or the nature of the sample (intact versus powdered).

#### 4.5. Establishment of equilibrium between sample and electrolyte

Most studies equilibrate the calcite–electrolyte systems of interest for 48 h or less prior to measuring the zeta potential (Table 1). Only a few studies have investigated longer equilibration times (e.g. [4,12,29,33,67]). Some studies have reported equilibration times of an hour or less (e.g. [11,18,48,58,60]). Heberling et al. [29] investigated the difference between equilibrium and non-equilibrium measurements. In a study using natural intact limestone, Alroudhani et al. [4] found that the equilibrium pH in open system conditions was reached after 100 h and sometimes more, and that it could take a factor of 2–3 times longer to reach the equilibrium pCa (e.g. Fig. 6). Thus there is the risk that short equilibration times may not yield a stable equilibrium in pH and/or pCa, which will affect the measured zeta potential. Moreover, the equilibrium pH may be achieved before the equilibrium pCa, so assessing equilibration based on pH measurements alone may not be sufficient. As discussed above, values of the zeta potential on calcite are primarily controlled by pCa and the concentration of other PDIs such as  $\text{Mg}^{2+}$ , rather than pH. Therefore, it is important to ensure that equilibrium in



**Fig. 6.** Calcite–water– $\text{CO}_2$  equilibrium. Plot (a) shows calcium concentration (expressed as pCa) and pH measured as a function of time during equilibration of natural Portland rock samples with deionized water. Plot (b) shows carbonate speciation into  $\text{H}_2\text{CO}_3$ ,  $\text{HCO}_3^-$ , and  $\text{CO}_3^{2-}$  as a function of pH. (Modified from Alroudhani et al. [4]).

the concentration of these PDIs has been reached prior to taking measurements. Inconsistent values of zeta potential obtained in some studies may reflect a lack of equilibration.

#### 4.6. Summary of published zeta potential data on calcite

Published measurements to date have typically observed linear, Nernstian relationships between zeta potential and pH, and between zeta potential and pCa, especially for values of pH and pCa close to the IEP. However, despite the clear relationship between zeta potential and pH obtained in such studies, the proton is not a PDI for calcite, as varying pH whilst holding pCa constant yields constant zeta potential within experimental error. Moreover, surface complexation models suggest that the calcite surface remains negatively charged over the pH range 5.5–11, and the surface charge is only weakly dependent on pH. The zeta potential is controlled instead by the concentration-dependent adsorption of lattice ions  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  in the Stern layer, and also by the adsorption of  $\text{Mg}^{2+}$  ions, although experimental data testing the effect of pMg are scarce. The few data available suggests that  $\text{Mg}^{2+}$  behaves identically to  $\text{Ca}^{2+}$  within experimental error. The  $\text{SO}_4^{2-}$  ion has also been suggested as a PDI for the calcite surface, but the evidence is not conclusive.

A wide range of values of IEP (expressed as pCa) have been observed, depending upon whether the sample is natural or artificial, whether the experiments are open or closed to atmospheric  $\text{CO}_2$ , and the approach taken to equilibrate the sample and electrolyte prior to obtaining data. In open system experiments, measured  $\text{pCa}_{\text{IEP}}$  lies in the range of 4.25–4.5, but in closed system experiments,  $\text{pCa}_{\text{IEP}}$  has a wider range of 0.4–4 and is always lower than values reported in open system experiments. Natural samples generally show more negative values of zeta potential than artificial samples for a given value of pCa, and therefore yield lower values of  $\text{pCa}_{\text{IEP}}$ . Recent studies of zeta potential on natural carbonates in closed system experiments consistently show the lowest values of  $\text{pCa}_{\text{IEP}}$  in the range 0.4–0.75.

### 5. Measurements of zeta potential on natural limestone rock samples: Impact of rock type, PDI concentration and ionic strength

As outlined in the previous section, few studies have reported measurements of zeta potential in carbonates at conditions relevant to natural subsurface systems. Most have explored synthetic calcite and dilute electrolytes with much lower total ionic strength and PDI concentration than subsurface brines. Moreover, most did not employ an experimental method that established equilibrium conditions of pH,  $\text{pCO}_2$  and PDI concentration relevant to subsurface carbonates. The aim of this section is to report new measurements of zeta potential relevant to natural systems, with a particular emphasis on determining (i) the relationship between pH and pPDI ( $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ ) in equilibrated calcite–electrolyte systems, given that no studies have measured pPDI whilst varying pH, or reported pH whilst varying pPDI, (ii) the effect of high concentrations of indifferent ions such as NaCl on the sensitivity of the zeta potential to

pPDI, (iii) whether  $\text{SO}_4^{2-}$  is a PDI for the calcite mineral, and (iv) how the zeta potential is affected by pPDI over the concentration range found in natural brines. Only Alroudhani et al. [4] have probed this parameter space, and they used only a single rock type. Here we investigate two further natural carbonates to determine whether the results are rock-type specific.

We used the SPM described by Jackson and co-workers [4,35,38,39,83] to measure the zeta potential. The SPM is applicable to intact natural samples, can be used to measure zeta potential at high ionic strength ( $>2\text{ M}$ ), and can also be used during multiphase flow and at elevated temperature [1,2,4,35,37,38,83,84]. The SPM measurements were complemented by chemical analysis of the effluent electrolyte, to monitor adsorption/desorption of ionic species during the experiments. The results have broad application to earth engineering processes such as hydrocarbon production and geological  $\text{CO}_2$  storage, and we use them here to investigate the mechanisms that underpin the use of controlled injection brine compositions during waterflooding of hydrocarbon reservoirs [41,74,86,87].

#### 5.1. Materials and experimental method

We used two intact natural carbonate core samples that in x-ray diffraction (XRD) analysis appear to be identical pure limestone that differ only in age and permeability (Table 2). Also shown in Table 2 are the properties of the Portland Limestone samples investigated by Alroudhani et al. [4], which we compare to the new results presented here when possible. Samples were cleaned using the enhanced method reported in Alroudhani et al. [4]. The brines used were synthetic solutions of reagent-grade NaCl,  $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ ,  $\text{Na}_2\text{SO}_4$  and  $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$  salts (Sigma Aldrich, UK) in deionized water (DIW) from a filtered system with electrical conductivity below  $1\text{ }\mu\text{S/cm}$ .

The equilibrium condition of carbonate/electrolyte/ $\text{CO}_2$  discussed previously (see Fig. 6) was replicated following the approach of Alroudhani et al. [4]. NaCl electrolytes of varying concentrations (0.05–5 M) were equilibrated by leaving offcuts of the tested carbonate samples in a closed beaker containing an air gap to allow  $\text{CO}_2$  to dissolve into solution, replicating the open-system conditions of carbonate deposition [76]. The pH of the NaCl electrolyte (measured using a Five-Go Mettler-Toledo pH meter with their 3-in-1 pH electrode LE438, implementing where necessary the manufacturer's recommended calibration and correction procedures at high ionic strength) and  $\text{Ca}^{2+}$  concentration (measured using Inductively Coupled Plasma Atomic Emission Spectroscopy, ICP-AES) were monitored until both reached constant value within experimental error. The equilibrium pH was found to be  $8.3 \pm 0.1$  for both Ketton and Estailades, in agreement with Alroudhani et al. [4], and the equilibrium pCa was found to be  $2.8 \pm 0.1$  and  $3.1 \pm 0.1$ , respectively. The equilibrated NaCl solutions were termed NaCl-EQ and were then used directly in zeta potential measurements, or were modified by addition of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and/or  $\text{SO}_4^{2-}$ .

The apparatus used to measure zeta potential in our SPM is closed to the atmosphere, and the second stage of equilibration prior to measuring

**Table 2**  
Properties and mineralogy of rock samples used in this study compared to Portland used by Alroudhani et al. [4].

| Property/rock                  | Ketton                                                           | Estailades                                                       | Portland                                                         |
|--------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| Description                    | Middle Jurassic Oolitic limestone from UK                        | Upper Cretaceous limestone from France                           | Upper Jurassic Oolitic limestone from UK                         |
| Porosity                       | $23\% \pm 0.5$                                                   | $28\% \pm 0.5$                                                   | $20\% \pm 0.5$                                                   |
| Permeability                   | $1.4\text{ Darcy} \pm 0.4$                                       | $0.13\text{ Darcy} \pm 0.2$                                      | $0.005\text{ Darcy} \pm 0.001$                                   |
| Intrinsic formation factor (F) | $13.87 \pm 0.5$                                                  | $12.92 \pm 0.5$                                                  | $22.04 \pm 0.5$                                                  |
| Compositions                   | 97% calcite ( $\text{CaCO}_3$ )<br>3% magnesium*                 | 97% calcite ( $\text{CaCO}_3$ )<br>3% magnesium*                 | 96.6% calcite ( $\text{CaCO}_3$ )<br>3.4% quartz                 |
| Dimensions                     | Length (L) = $0.076\text{ m}$<br>Diameter (D) = $0.038\text{ m}$ | Length (L) = $0.076\text{ m}$<br>Diameter (D) = $0.038\text{ m}$ | Length (L) = $0.076\text{ m}$<br>Diameter (D) = $0.038\text{ m}$ |

\* Magnesium ( $\text{Mg}^{2+}$ ) is likely incorporated into the calcite structure as  $\text{MgCO}_3$  or  $\text{CaMg}(\text{CO}_3)_2$ .

the zeta potential was to ensure equilibrium between the electrolyte of interest (NaCl–EQ with or without the addition of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  or  $\text{SO}_4^{2-}$ ) and the rock sample at the closed-system conditions pertaining to a rock–brine system at depth. The rock sample was pre-saturated with the selected electrolyte at open-system conditions and then confined in the core holder at closed-system conditions, and the electrolyte was pumped through the sample from the (closed) inlet reservoir to the (closed) outlet reservoir and back again. The repeated flow of the electrolyte through the sample at closed system conditions mimics migration of the electrolyte into the carbonate rock at depth. At regular intervals, the electrical conductivity and pH of the electrolyte in the reservoirs was measured, and equilibrium was assumed to have been reached when the conductivity and pH of the electrolyte in each reservoir differed by <5%.

Electrolyte samples were analysed before and after the SPM experiments for key ions ( $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$ ). We used ICP-AES for cations and Ion Chromatography (IC) for anions with appropriate dilution where necessary. The ICP-EAS analysis was carried out in the Analytical Chemistry Laboratory at the Natural History Museum London; the IC analysis was carried out in the TOTAL Laboratory for Reservoir Physics at Imperial College London. Instrument error was determined by using certified solutions containing  $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  and the error calculated from the standard deviation of five repeat measurements of each solution.

## 5.2. Design of experiments

We conducted two sets of experiments. In the first, the sensitivity of zeta potential to variations in NaCl concentration was tested using the equilibrated NaCl–EQ solutions described above. In these experiments, the concentration of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  was dictated by the equilibration process; no additional salts containing these ionic species were added to the NaCl–EQ solutions. In the second, the sensitivity of

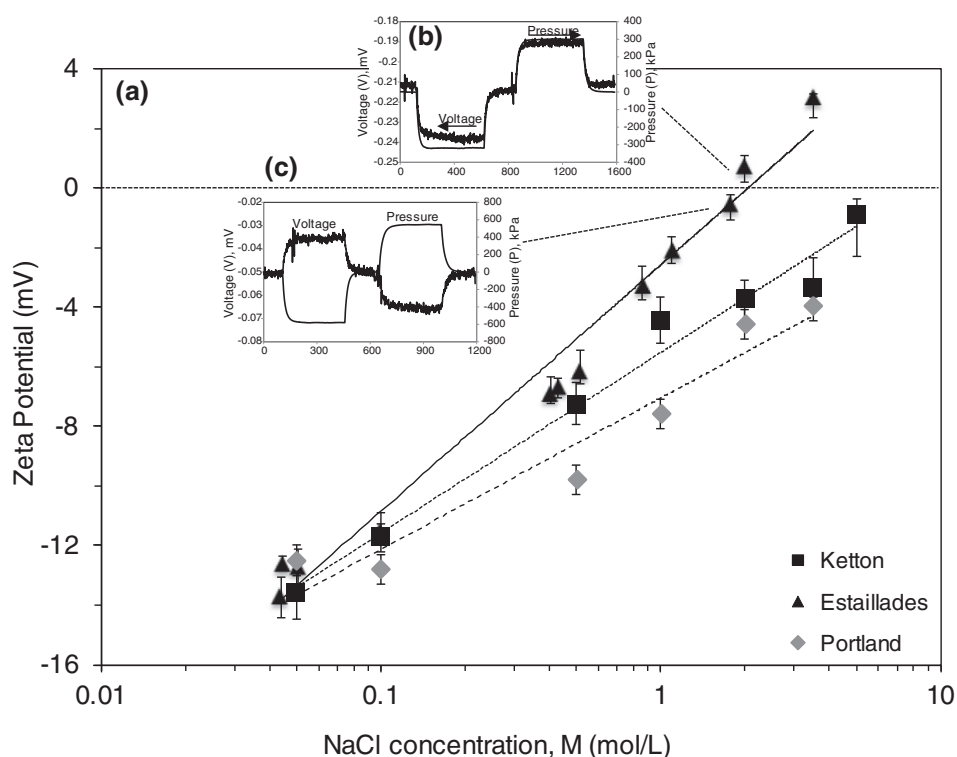
the zeta potential to variations in  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  concentration was tested by adding salts containing these ionic species over the range found in natural brines (0.007–0.42 M for  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ ; 0.002–0.096 M for  $\text{SO}_4^{2-}$ ) to NaCl–EQ solutions of two different ionic strengths (0.5 M and 2 M). The 0.5 M NaCl concentration represents seawater and is similar to the ‘ZP brine’ of Zhang and Austad [88] and Zhang et al. [87], allowing direct comparison of results. The 2 M NaCl concentration represents the saline brines found in many saline aquifers and hydrocarbon reservoirs (e.g. [59]). In the second set of experiments, we investigated for the first time the relationship between pCa (or pMg) and pH. Note that in the experiments reported here, precipitation of salts such as  $\text{CaSO}_4$  and  $\text{MgCO}_3$  in the pore-space was prevented because each ion of interest ( $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  or  $\text{SO}_4^{2-}$ ) was added to NaCl–EQ electrolyte containing only trace or zero concentration of cations or anions other than  $\text{Na}^+$  and  $\text{Cl}^-$ .

## 5.3. Results

Streaming potentials were measured using the paired-stabilization method of Vinogradov et al. [83] and typical raw results are shown in Fig. 7b,c. The stabilized voltage was plotted against the stabilized pressure difference across the sample for four different flow rates and a linear regression through the data yielded the streaming potential coupling coefficient and, via Eq. (3), the zeta potential. The uncertainty in the streaming potential coupling coefficient arising from the range of linear regressions that can be forced through the stabilized voltage and pressure data was used to determine the associated experimental error in zeta potential reported in the following sections.

### 5.3.1. Impact of $\text{Na}^+$ concentration on zeta potential

We begin by reporting the effect of NaCl concentration on zeta potential. When possible, we compare results from the Ketton and

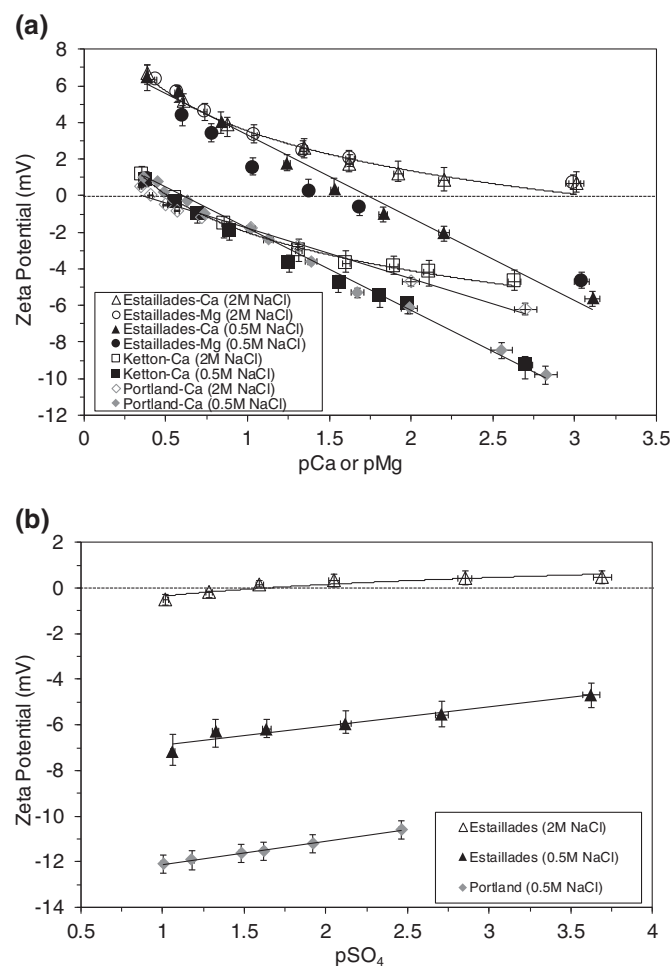


**Fig. 7.** (a) Effect of NaCl concentration on zeta potential on the Ketton (squares) and Estailades (triangles) samples. Data obtained on the Portland carbonate sample by Alroudhani et al. [4] are shown for comparison (grey diamonds). The inset plots (b) and (c) show typical example measurements of voltage and pressure against time from the SPM used to determine the zeta potential. When pressure and voltage respond in the same sense the zeta potential is positive and vice-versa. The polarity of the zeta potential is accurately determined even when its value is close to zero.



Estailades samples investigated here against the Portland sample studied by Alroudhan et al. [4]. Fig. 7a shows the zeta potential as a function of NaCl concentration (in M) on a log-log scale for all three natural carbonate samples. A linear regression can be fitted to the data from each rock type (with  $R^2 > 0.97$ ) but the gradient of the regression differs between samples. The Estailades sample (filled triangles) shows the highest gradient ( $-8.08 \pm 0.52$  mV/M) and the Portland sample (filled grey diamonds) the lowest gradient ( $-5.08 \pm 0.47$  mV/M) with the Ketton sample (filled squares) in between ( $-6.11 \pm 0.49$  mV/M).

At low NaCl concentration ( $<0.1$  M) the zeta potential is negative and identical (within experimental error) for all three samples. However, as NaCl concentration increases, the zeta potentials diverge with  $|\zeta_{\text{Portland}}| > |\zeta_{\text{Ketton}}| > |\zeta_{\text{Estailades}}|$ . Moreover, the zeta potential of the Estailades samples becomes positive at NaCl concentration  $> 1.9$  M, whereas the other two samples yield negative zeta potential over the entire concentration range investigated. The inset plots 7b and 7c show there is no ambiguity in the polarity of the zeta potential: for a NaCl concentration of 1.8 M the streaming potential, and hence zeta potential, of the Estailades sample is clearly negative ( $-0.56$  mV  $\pm$  0.33) and becomes positive ( $0.72$  mV  $\pm$  0.33) for a NaCl concentration of 2 M. As discussed above,  $\text{Na}^+$  and  $\text{Cl}^-$  are believed to be indifferent ions to the calcite mineral surface and their presence is expected to affect the magnitude, but not the polarity, of the zeta potential (e.g. [23,34,54]).



**Fig. 8.** Zeta potential versus (a) pCa and pMg, and (b) pSO<sub>4</sub> for the Ketton (squares) and Estailades (triangles and circles) samples at different NaCl concentrations. Data obtained on the Portland carbonate sample by Alroudhan et al. [4] are shown for comparison (diamonds).

### 5.3.2. Impact of $\text{pCa}^{2+}$ , $\text{pMg}^{2+}$ and $\text{pSO}_4^{2-}$ on zeta potential

We next investigate the impact of the concentration of the confirmed and suggested PDIs  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{SO}_4^{2-}$  on the zeta potential. We plot concentration as pPDI and, in all cases, the highest pPDI values reported correspond to the equilibrium concentrations in the NaCl-EQ electrolytes. Fig. 8a shows the impact of pCa and pMg on the zeta potential of Estailades, and the impact of pCa on the zeta potential of Ketton, for two different NaCl concentrations (0.5 M and 2 M) typical of subsurface brines; also shown for comparison are the data from Portland obtained by Alroudhan et al. [4]. We find that the zeta potential consistently becomes more positive with decreasing pCa or pMg. Moreover, the response to pCa and pMg is identical within experimental error for the Estailades carbonate, irrespective of NaCl concentration. Alroudhan et al. [4] also found that pCa and pMg behaved identically for the Portland carbonate, although they investigated a lower NaCl concentration of 0.05 M and their data are not shown here.

The response of the zeta potential to pCa or pMg is generally sample specific and depends on the NaCl concentration. At the lower (0.5 M) NaCl concentration investigated here, Ketton and Portland exhibit identical behaviour over the entire pCa range investigated: a linear regression provides an excellent fit to the data ( $R^2 > 0.98$ ) with a gradient of  $-4.5 \pm 0.25$  mV/decade and an  $\text{pCa}_{\text{IEP}}$  of  $0.5 \pm 0.03$ . A linear regression with the same gradient also provides an excellent fit to the Estailades data at the lower NaCl concentration investigated; however,  $\text{pCa}_{\text{IEP}}$  is much higher at  $1.7 \pm 0.03$ . Nernstian (linear) behaviour is therefore always observed at the lower NaCl concentration over the entire pCa and pMg range investigated, but the IEP is different for the Estailades sample. Fitting the Gouy–Chapman–Grahame model (Eq. 18) to these data yields  $C_s = 1 \text{ Fm}^{-2}$  and  $\Delta = 0.245 \text{ nm}$ , similar to the values obtained for Portland carbonate by Alroudhan et al. [4] and within the range used by Heberling et al. [29].

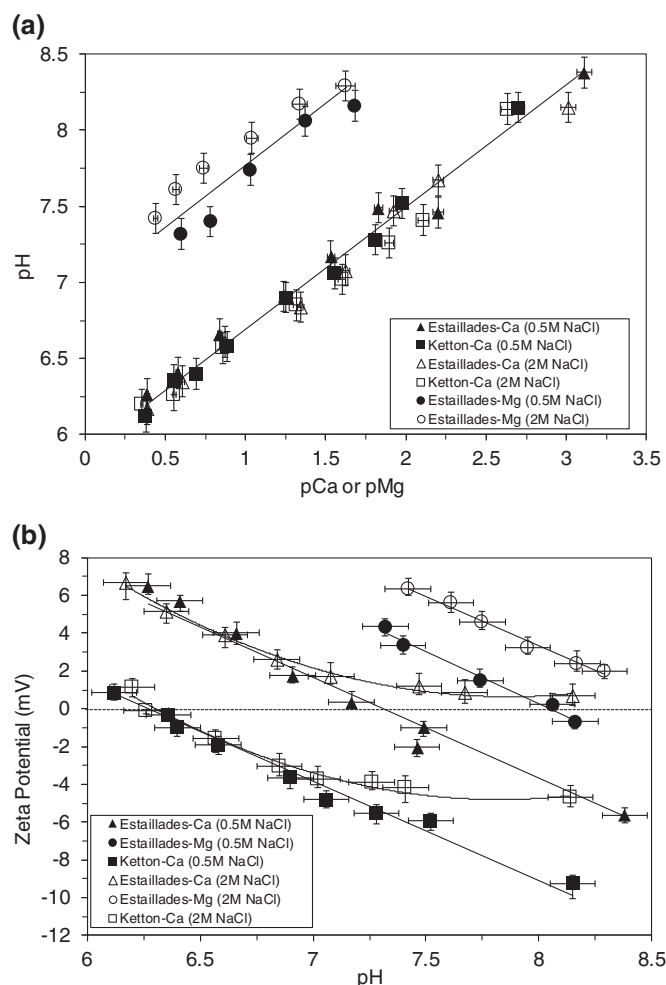
At the higher (2 M) NaCl concentration investigated here, the Ketton carbonate continues to show linear, Nernstian behaviour close to the IEP with a gradient and IEP identical to that obtained using 0.5 M NaCl within experimental error, but the behaviour becomes non-linear (non-Nernstian) away from the IEP at high pCa. The Portland sample shows linear, Nernstian behaviour over the entire pCa range investigated with an identical IEP within experimental error but a lower gradient of  $-2.86 \pm 0.25$  mV/decade. The Estailades carbonate exhibits notably different behaviour, with the zeta potential remaining positive over the entire pCa/pMg range investigated, and non-Nernstian behaviour at high pCa. Cicerone et al. [12] also observed non-Nernstian behaviour far from IEP and related that to variations in the Stern layer capacitance or to breakdown of the Gouy–Chapman–Grahame model.

Increasing pSO<sub>4</sub> consistently yields more positive zeta potential, although the sensitivity is much less than that observed when varying pCa or pMg (Fig. 8b). A linear regression yields a good fit to the data at the lower NaCl concentration investigated here ( $R^2 > 0.92$ ) with a consistent gradient of  $0.84 \pm 0.2$  mV/decade for both the Estailades and Portland carbonates. A key feature is that an apparent IEP of  $\text{pSO}_4 = 1.59 \pm 0.2$  is observed for the Estailades sample in 2 M NaCl electrolyte. It is also notable that the difference in absolute zeta potential between the different carbonate samples observed in Fig. 7 persists in Fig. 8, with  $|\zeta_{\text{Portland}}| > |\zeta_{\text{Ketton}}| > |\zeta_{\text{Estailades}}|$ .

### 5.3.3. Relationship between pH and pCa or pMg

Fig. 9a shows the relationship between the equilibrium pH and pCa or pMg obtained across all of the experiments reported here. A linear regression provides an excellent fit to the data ( $R^2 > 0.98$ ) with a gradient of 0.81, irrespective of NaCl concentration, rock sample or PDI varied, although the value of pH is consistently higher in the experiments varying pMg as compared to those varying pCa. These experimental data confirm that equilibrium pH and pPDI are related at fixed  $\text{pCO}_2$  via Eqs. (14)–(17). It is possible to plot the data obtained here against pH and obtain an apparently strong correlation (Fig. 9b) even though pCa and pMg were the controlled parameters. The





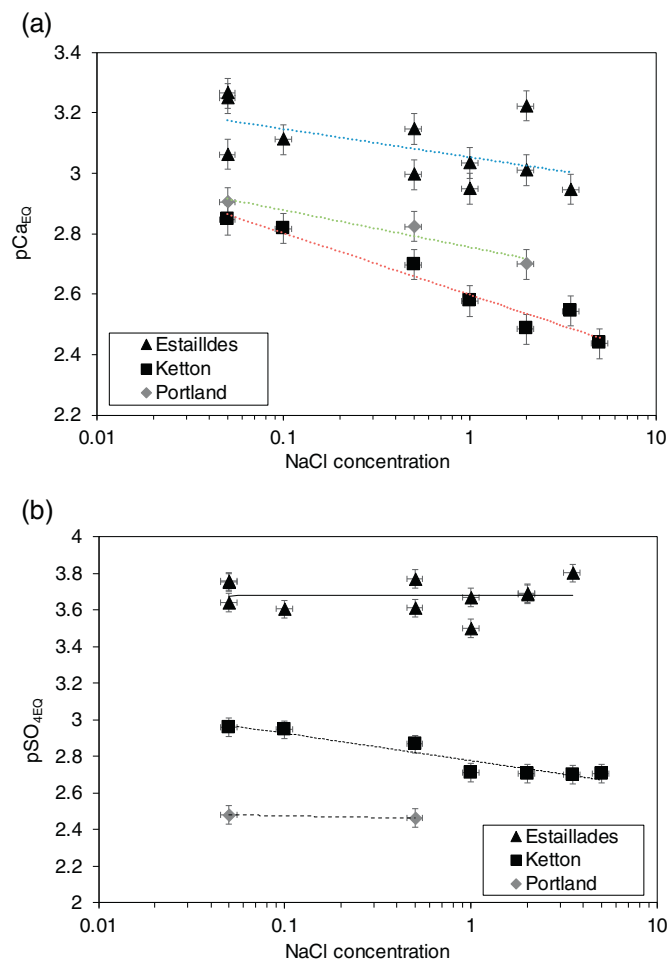
**Fig. 9.** (a) Relationship between pH and pCa or pMg obtained from all the experiments shown in Fig. 8, along with linear regressions to the data. (b) Data shown in Fig. 8 replotted against pH using the linear regression shown in (a).

relation between pH and pCa or pMg obtained here is consistent with model predictions (e.g. PHREEQC; [55]). However, it has not previously been demonstrated in properly equilibrated experiments where zeta potential was also measured, or over such a wide range of pCa.

#### 5.4. Discussion

##### 5.4.1. Impact of indifferent ions

The results presented here show a clear relationship between the concentration of indifferent ions  $\text{Na}^+$  and  $\text{Cl}^-$  and the zeta potential on natural carbonate (Fig. 7). Such a relationship is expected owing to collapse of the double layer with increasing ionic strength, but if double layer collapse is the only control on zeta potential, then the zeta potential should be proportional to the logarithm of NaCl concentration (as a proxy for ionic strength; e.g. [27]). Consequently, the dependence of zeta potential on NaCl concentration observed in Fig. 7 cannot result only from double layer collapse. Fig. 10 shows the equilibrium pCa and  $\text{pSO}_4$  as a function of NaCl concentration, corresponding to the zeta potential data shown in Fig. 7. Note that pMg remained large and approximately constant at  $4.19 \pm 0.03$  for all samples irrespective of NaCl concentration (i.e. the  $\text{Mg}^{2+}$  concentration is small). pCa generally decreases with increasing NaCl concentration (Fig. 10a); hence, given that zeta potential becomes more positive with decreasing pCa, these data are consistent with the more positive zeta potential observed



**Fig. 10.** Equilibrium pCa (a) and  $\text{pSO}_4$  (b) versus NaCl concentration for the Ketton (squares) and Estailades (triangles) samples. Data obtained on the Portland carbonate sample by Alroudhani et al. [4] are shown for comparison (diamonds).

with increasing NaCl concentration in Fig. 7. However, the sensitivity decreases at high NaCl concentration, moreover, these data do not explain the polarity change of the zeta potential observed for the Estailades carbonate. This remains a curious feature of the new data obtained here and may suggest that  $\text{Na}^+$  is not in fact an indifferent ion to the calcite surface in some circumstances, as suggested by [29,44].

##### 5.4.2. Impact of carbonate type

The results obtained here also show that natural carbonate samples that appear identical in XRD display different zeta potential behaviour. The Estailades carbonate, in particular, behaves differently to the Ketton and Portland carbonates, showing inversion of the zeta potential with increasing NaCl concentration (Fig. 7), higher values of  $\text{pCa}_{\text{IEP}}/\text{pMg}_{\text{IEP}}$  compared to Ketton and Portland (Fig. 8a), and also an apparent IEP expressed as  $\text{pSO}_4$  (Fig. 8b). The Estailades carbonate released the least  $\text{SO}_4^{2-}$  during equilibration with the NaCl solutions, while the Portland sample released the most  $\text{SO}_4^{2-}$  (Fig. 10b), consistent with the observed differences in zeta potential ( $\zeta_{\text{Portland}} > \zeta_{\text{Ketton}} > \zeta_{\text{Estailades}}$ ) in Figs. 7 and 8. We hypothesize that Ketton and Portland contain undetected sources of  $\text{SO}_4^{2-}$  which affects the pristine zeta potential, yielding more negative values and lower  $\text{pCa}_{\text{IEP}}$  than Estailades. Natural carbonates can incorporate a number of different minerals, including clays, organic matter, apatite, anhydrite or quartz, that yield a more negative zeta potential than pure calcite [12,65,66]. Our data therefore suggest that the behaviour of the natural Ketton and Portland carbonate samples investigated here is affected by the presence of impurities such as

anhydrite. In contrast, the behaviour of the Estailades sample is consistent with the ionic strength effect as calculated with a Pitzer model for a calcite-atmospheric  $\text{CO}_2$  equilibrium solution [55].

The contrasting behaviour of Estailades carbonate may also reflect differences in the affinity of  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  for the mineral surface. Pierre et al. [54] suggested that the IEP is governed by the relative magnitude of the equilibrium constants  $K_{\text{Ca}}$  and  $K_{\text{CO}_3}$  governing the adsorption of  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$  ions on the calcite mineral surface. The IEP shifts to lower pCa if  $K_{\text{CO}_3} > K_{\text{Ca}}$ ; that is, if the calcite surfaces show greater affinity for  $\text{CO}_3^{2-}$  than  $\text{Ca}^{2+}$ . Pierre et al. [54] found the IEP differed for synthetic and natural calcite and argued that this reflected the differing affinity for Ca and  $\text{CO}_3$ . The Pierre et al. model suggests that the Estailades carbonate investigated here has a much lower affinity to  $\text{CO}_3^{2-}$  (and greater affinity to  $\text{Ca}^{2+}$ ) than the Ketton and Portland carbonates at equilibrium conditions.

#### 5.4.3. $\text{SO}_4$ as a PDI for calcite

Several studies have suggested that the  $\text{SO}_4^{2-}$  ion is a PDI for calcite (e.g. [64,88]). However, these studies did not measure changes in pCa or pMg in response to changing p $\text{SO}_4$ . Here, we find that  $\text{SO}_4^{2-}$  has only a minor influence on the zeta potential of natural carbonate (Fig. 8b). At low ionic strength (0.5 M NaCl), the change in p $\text{SO}_4$  correlates to a change in pCa (Fig. 11a) which can explain the change in zeta potential (Fig. 11b). We observe a linear relationship between zeta

potential and pCa with a similar gradient (5.1 mV/decade;  $R^2 > 0.85$ ) to that obtained in the experiments where pCa was varied (4.5 mV/decade). However, at high ionic strength (2 M NaCl), pCa is relatively insensitive to p $\text{SO}_4$  (Fig. 11a), yet the zeta potential still varies in response to changing p $\text{SO}_4$  and changes polarity from positive to negative. Moreover, there is not a clear correlation between pCa and zeta potential (Fig. 11b). It may be that small changes in pCa, pMg and even pNa within experimental error all contribute to the observed change in zeta potential, as it is only c. 1 mV in magnitude and close to the IEP. Geochemical modelling beyond the scope of the paper is required to test this hypothesis.

#### 5.4.4. Implications for wettability alteration and hydrocarbon recovery in carbonate reservoirs

In carbonate rock formations with positively charged mineral surfaces, an attractive electrostatic force will act between the mineral surfaces and the negatively charged oil-brine interface, promoting wettability alteration to oil-wet conditions (e.g. [10,31]). Several studies have suggested that carbonate rocks saturated with natural brines rich in  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  ions have positively charged mineral surfaces so are likely to be oil-wet (e.g. [4,11,38,46,47]). The wettability of a reservoir plays a key role in the efficiency with which the oil can be produced. However, we find here considerable variability in the IEP expressed as pCa, suggesting that some carbonates will be positively charged in contact with natural brines, and others will not. The SPM used here is a suitable technique to determine the zeta potential of intact reservoir rock samples saturated with natural brine (e.g. [38]).

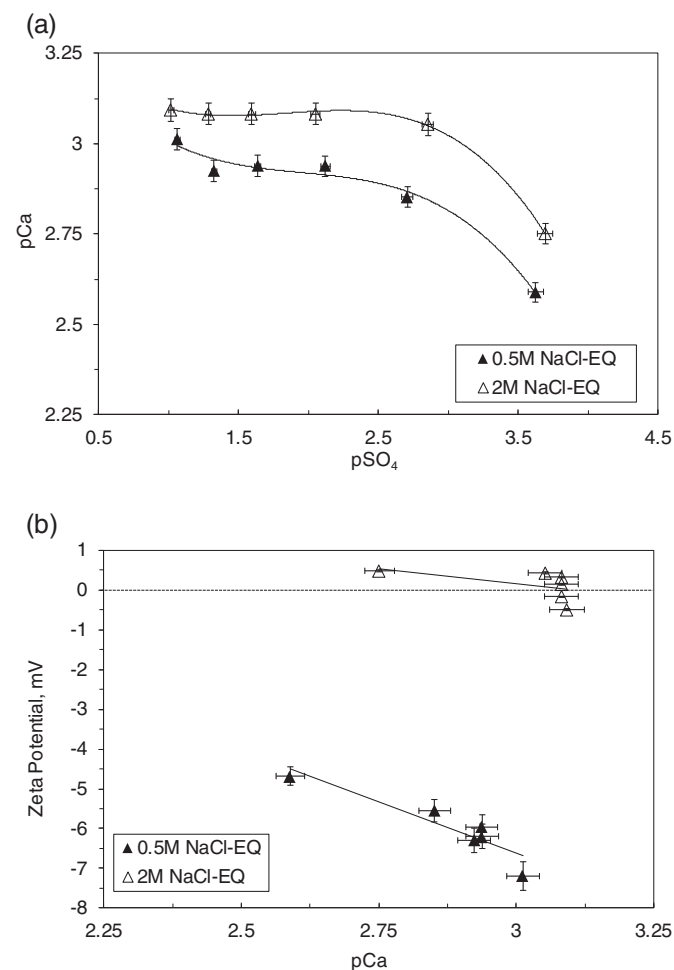
In many reservoirs, water is injected to maintain the pressure and displace oil towards production wells. If the concentration of  $\text{Ca}^{2+}$  or  $\text{Mg}^{2+}$  in the injected water can be modified to fall below the IEP, either selectively or by bulk dilution in a process termed ‘controlled salinity waterflooding’ (CSW), the zeta potential will change polarity from positive to negative. The polarity change leads to electrostatic repulsion between the mineral surfaces and the negatively charged oil-brine interface. This may promote wettability alteration to water-wet conditions, releasing previously adsorbed oil from the calcite mineral surfaces and therefore improving oil recovery (e.g. [47]).

Most previous reported values of the IEP expressed as pCa suggest that considerable reduction in Ca concentration is required to change the polarity of calcite (Table 1). However, our results suggest that far less dilution may be required in some natural carbonates (e.g. Ketton and Portland); in others, the IEP may never be encountered and the mineral surfaces may remain positively charged (e.g. Estailades). The carbonate-rock-specific relationship between zeta potential and PDI concentration observed here may explain why CSW in some cases yields increased recovery and in other cases does not (e.g. [3,59]). Future work relevant to CSW should focus on testing the link between brine composition, zeta potential and increased oil recovery using integrated experiments with consistent materials and experimental conditions (e.g. [37]).

## 6. Conclusions

Experimental studies of the zeta potential of calcite and carbonate have reported a broad range of values with numerous inconsistent results. Yet the primary data source to understand calcite surface charge comprises measurements of zeta potential, so it is important to understand the key controls on zeta potential and why studies have reported apparently contradictory values. A comprehensive review of the literature, supplemented by experimental data new to this study, suggests that:

1. The potential determining ions (PDIs) for the carbonate mineral surface are the lattice ions  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and  $\text{CO}_3^{2-}$ . The zeta potential is controlled by the concentration-dependent adsorption of these ions within the Stern layer, primarily at the Outer Helmholtz Plane (OHP).



**Fig. 11.** Impact of p $\text{SO}_4$  on pCa and zeta potential measured on the Estailades sample. (a) pCa measured during the experiments shown in Fig. 8b in which  $\text{SO}_4^{2-}$  was added to pre-equilibrated NaCl brine plotted against measured p $\text{SO}_4$ . (b) Zeta potential from the experiments shown in Fig. 8b plotted against the measured pCa recorded in (a).

- Given (1), the Iso-Electric Point (IEP) at which the zeta potential is zero should be expressed as pCa (or pMg). It should not be reported as pH, similar to most metal oxides.
- The pH does not directly control the zeta potential. Varying the pH whilst holding pCa constant yields constant zeta potential. The pH affects the zeta potential only by moderating the equilibrium pCa for a given CO<sub>2</sub> partial pressure (pCO<sub>2</sub>).
- Experimental studies that appear to yield a systematic relationship between pH and zeta potential are most likely observing the relationship between pCa and zeta potential, with pCa responding to the change in pH. New experimental data presented here show a consistent linear relationship between equilibrium pH and equilibrium pCa or pMg irrespective of sample or solution ionic strength.
- The surface charge is weakly dependent on pH through protonation and deprotonation reactions that occur within a hydrolysis layer immediately adjacent to the mineral surface. The Point of Zero Charge (PZC) at which the surface charge is zero could be expressed as pH, but surface complexation models suggest the surface is negatively charged over the pH range 5.5–11.
- Several studies have suggested that SO<sub>4</sub><sup>2−</sup> is also a PDI for the calcite surface, but new experimental data presented here indicate that the value of pSO<sub>4</sub> may affect zeta potential only by moderating the equilibrium pCa. Modelling will be required to provide a more thorough interpretation of the new data.
- Natural carbonate typically yields a more negative zeta potential than synthetic calcite, most likely due to the presence of impurities including clays, organic matter, apatite, anhydrite or quartz, that yield a more negative zeta potential than pure calcite. New data presented here show that apparently identical natural carbonates display differing zeta potential behaviour, most likely due to the presence of small volumes of these impurities.
- It is important to ensure that equilibrium, defined in terms of the concentration of PDIs, has been reached prior to taking measurements. Inconsistent values of zeta potential obtained in some studies may reflect a lack of equilibration.
- The data collated and reported here have broad application in engineering processes including the manufacture of paper and cement, the geologic storage of nuclear waste and CO<sub>2</sub>, and the production of oil and gas.

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